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DEPARTMENT OF  
**STATISTICS**

# Learning Optimal Commodity Taxation

by

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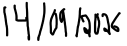
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This is my own work (except where otherwise indicated)

I used ChatGPT 5.6 Sol for editing and formatting. I used that same model's Pro version for guidance on a number of results related to the minimax lower bound and the rank factorization theorem.

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## Abstract

I study an optimal commodity taxation problem where a policymaker seeks to maximize social welfare—sales tax revenue less the loss in private household welfare from increased prices. In my formulation, the policymaker learns a social welfare maximizing taxation policy of commodities when experimentation does not directly reveal social welfare under each candidate taxation. To find the optimal taxation, I design a sequential experiment called *Bregman Contrast Pruning*: an algorithm that organizes the continuous tax domain into a hierarchical partition, eliminates tax regions that cannot contain an optimum and refines only the surviving regions. Under subgaussian observation noise, the procedure is  $(\varepsilon, \delta)$ -correct: with probability at least  $1 - \delta$ , it returns a taxation whose welfare is within  $\varepsilon$  of maximum social welfare. A key feature of Bregman Contrast Pruning is instance-dependent regret. The algorithm connects the economically induced curvature of demand to the number and placement of samples required to learn an approximately optimal commodity taxation, avoiding full estimation of the entire social welfare field.

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# 1 Introduction

I study the optimal taxation of a set of commodities and design an experiment where a policymaker learns the welfare-maximizing vector of commodity taxes (called the *optimal taxation*), where household (consumers) demand is heterogeneous and initially unknown. The problem of the optimal taxation of commodities as first introduced by Frank Ramsey asks how taxes should be distributed across commodities when the policymaker wants to maximize so-called *social welfare* (Cesa-Bianchi et al., 2025). Here social welfare is sales tax revenue less the loss of consumer welfare caused by increased costs (households must pay higher prices and are thus worse off). Ramsey’s analysis lead to the well-known Ramsey Tax rule, which states that commodities with relatively inelastic (invariable with respect to price) demand ought to have higher taxes because their sales respond less strongly to a price increase (Ramsey, 1927). With several goods, however, cross-commodity substitution (correlated demand) makes the problem more difficult. A tax on one commodity may change demand for many others, so the welfare effect of a tax vector cannot generally be determined by such a calculation.

Classical optimal-tax calculations treat the relevant demand system as known. In empirical applications, it is not. A policymaker may observe household purchases under historical taxes, but population demand, cross-tax substitution responses (change in demand due to a change in tax), and consumer surplus (consumer’s willingness to pay less what they must pay) must be inferred from those observations (Mirrlees, 1971; Chetty, 2009). I suppose (possibly noisy) household purchasing behavior is observed under each taxation via a sequential experiment. At each sampling round, the policymaker assigns a common candidate commodity taxation (vector of taxes) to a fresh cohort (every household is subjected to the same taxes) sampled independently from a stable household population. The purpose of each candidate taxation is not to maximize social welfare during the experiment itself, but to learn an optimal taxation that will subsequently be implemented. The experiment then ends after a finite number of sampling rounds, when the policymaker recommends one taxation and that recommendation is evaluated by the difference between the social welfare under it and the true maximum social welfare (simple regret).

Finding the optimal taxation of commodities through a sequential experiment is an instance of a broader indirect-feedback pure-exploration problem, as studied in the literature on bandits. Classical work on best-arm (in this case optimal taxation) identification ordinarily begins from direct feedback: after choosing an arm (e.x. candidate taxation), the learner observes a random reward (what the learner wants to maximize) generated by that arm (Robbins, 1952; Bubeck et al., 2011a; Garivier and Kaufmann, 2016; Kasy and Sautmann, 2021). Many policy problems, including those beyond taxation, do not have this form. In dose-finding studies, investigators may observe toxicity, efficacy, or other biomarkers that are informative about clinical benefit without directly measuring that benefit (Villar et al., 2015; Pallmann et al., 2018; Aziz et al., 2021; Berry, 2006; Berry et al., 2010; U.S. Food and Drug Administration, 2026). Optimal taxation provides another central example. At each sampling round, purchase outcomes are observed (although they are possibly noisy), whereas the actual reward, social welfare, is not (Mirrlees, 1971; Chetty, 2009; Cesa-Bianchi et al., 2025).

The direct vs in-direct feedback distinction matters because maximizing an observed be-

havioral response, such as purchases, need not maximize social welfare. A taxation that raises substantial revenue may impose a large loss of consumer welfare. Conversely, a policy that preserves consumption may collect too little revenue. The observed response is an input into the welfare calculation, not the objective itself. A learning procedure must therefore combine behavioral observations with economic structure before the policies can be ranked.

I impose a general but structured economic setting in which household preferences are represented by *quasilinear* utility functions without income effects, i.e. utility is linear in a numeraire good (e.x. money) and possibly nonlinear in every other commodity. The model also permits heterogeneous priority weights for distinct groups of households within the population. A policymaker may place greater weight on the welfare of low-income households, on groups that bear a disproportionate burden of taxation, their political supporters, or on any other group whose welfare receives special priority. Consideration of heterogeneous priority weights is related to the many-person Ramsey problem of Diamond (1975) and the generalized social marginal welfare weights of Saez and Stantcheva (2016). Heterogeneous priority weights distinguish an ordinary social welfare objective from a weighted social welfare objective. Thus, two taxations that generate similar aggregate purchases may have notably different consequences for the welfare function the policymaker intends to maximize.

As a simplified example, consider the following illustration. Imagine the medieval island kingdom of Sandwich. Its households purchase four commodities: bread, peanut butter, grape jelly, and hazelnut spread. By decree of their king, Bimbo I, Sandwich does not trade with any other kingdom. Households combine these goods to make either peanut-butter-and-jelly sandwiches or hazelnut-spread sandwiches. After years of pirate raids, Bimbo I must levy sales taxes to finance repairs to the castle. He must also avoid imposing excessive welfare losses on his subjects and, in particular, on the lords whose political support he values more heavily. Each commodity tax raises the consumer price of the good on which it is levied. A tax on peanut butter makes peanut butter-and-jelly sandwiches more expensive. Some households may substitute toward hazelnut spread, while demand for grape jelly may fall because grape jelly is complementary to peanut butter. A tax on one commodity can consequently change demand for several others. Households may also differ in their preferences. Lords may strongly prefer hazelnut spread, while commoners and peasants may be approximately indifferent between the two sandwich types. Bimbo I must account for these heterogeneous responses because tax revenue is collected only from completed purchases and because he assigns different social weights to the welfare of different groups.

Under the imposed economic setting, the ordinary demand geometry controls the extrapolation error of estimating social welfare under one taxation using data from samples collected under different taxation. When the curvature of demand is restricted to be quadratic, this geometry is a squared Mahalanobis geometry determined by a fixed substitution-response (covariance of demand with respect to tax) matrix. Without such a restriction, the substitution-response matrix varies across the tax domain, and the cost of transporting information across taxations depends on the direction and path of the comparison.

Using the curvature of demand to decide which candidate taxation to test yields the algorithm I call *Bregman Contrast Pruning* or BCP, which allows for the optimal taxation to be learned without first estimating the complete social welfare field. The procedure is consequently decision focused. Once a tax region is certified to be suboptimal, further estimation of welfare variation within that region cannot alter the final recommendation. BCP stops allocating samples to such regions. This distinguishes its objective from procedures whose intermediate goal is a uniformly accurate estimate of the social welfare field over the entire domain of candidate taxations.

## 1.1 Contributions

The first contribution is the geometric reduction of the welfare-recovery problem. I show that ordinary demand controls the first-order extrapolation error for weighted consumer surplus. I develop this relation separately for quadratic and nonquadratic demand structures. In the quadratic case, the geometry is Mahalanobis and rank deficiency restricts the directions in which the weighted demand can be nonlinear. In the nonquadratic case, the extrapolation error is an integrated directional curvature, and its asymmetry is controlled by the variation of the substitution-response matrix along an edge.

The second contribution is Bregman Contrast Pruning algorithm itself. The algorithm combines a continuous hierarchy of candidate taxation with directed pathwise welfare reconstruction. A common graph of candidate taxations allows observations at one tax to enter several policy comparisons. An independent pilot sample provides upper bounds on conditional covariance, and a convex allocation problem determines how many fresh samples should be assigned to each candidate taxation. The correctness theorem shows that social welfare maximizing taxations (within an  $\epsilon$ -neighborhood) are not eliminated with probability at least  $1 - \delta$ , while sufficiently suboptimal cells are pruned with the same probability. A matching upper and lower minimax regret bound over one exact-choice structural household class. Thus, for a fixed commodity dimension  $J$  and fixed cohort size  $m$ , the minimax expected simple regret satisfies  $T^{-1/(J+2)}$  decay up to logarithmic factors, meaning BCP is minimax-rate optimal up to logarithmic factors over the stated model class. Lastly, I show that BCP can take advantage of favorable demand curvature, improving the aforementioned regret decay under certain specified economic settings.

## 1.2 Related work

The economic problem I study and the statistical methods I employ stem from rich literature in both economics and statistics. I discuss some of the work related to optimal taxation, pure exploration, and experimental design below.

**Optimal taxation and welfare sufficient statistics.** Optimal Tax Theory ordinarily separates the normative and empirical parts of the policy problem. Priority weights and the social objective are specified theoretically; demand elasticities and other behavioral responses are estimated from historical or experimental variation; and the estimates are substituted into an optimal policy formula (Ramsey, 1927; Mirrlees, 1971; Diamond, 1975; Chetty, 2009; Saez and Stantcheva, 2016). The sufficient statistics approach emphasizes

that welfare consequences can often be recovered from a limited set of behavioral responses without estimating every primitive of a structural utility model (Chetty, 2009). I share this perspective but treat the collection of those behavioral statistics as a sequential design problem.

The optimal taxation problem in which I consider is most closely related to the one studied in Cesa-Bianchi et al. (2025), as adapted from Ramsey (1927). They study adaptive welfare maximization when a policymaker observes behavior but not private utility for in a single good market. In fact, they also sketch a multidimensional commodity tax extension but leave its formal regret characterization open. However, the principal objective of their designed experiment is to minimize cumulative regret instead of simple regret, and their baseline model has a scalar policy and a binary action. I study the commodity taxation problem directly, permit vector-valued continuous purchases, and evaluate the final recommendation through fixed confidence correctness and simple regret.

The discrete tax problem studied by Shao (2026) also provides a particularly close comparison. With finitely many tax rates and a positive welfare gap, the probability of recommending an incorrect tax, and hence expected simple regret, may decay exponentially. That being said, the continuous problem considered here is quite different. The model class contains policies with arbitrarily small welfare differences, and the learner must refine the location of the optimum as the experimental budget grows. This spatial approximation problem produces a polynomial rather than exponential worst-case rate.

**Pure exploration and policy choice.** Work on pure exploration with bandits studies how observations should be allocated when only the final recommendation is evaluated (Awerbuch et al., 1996; Bubeck et al., 2011a; Garivier and Kaufmann, 2016; Kaufmann and Koolen, 2021; Lattimore and Szepesvári, 2020). Fixed confidence procedures seek a correct recommendation with a prescribed probability while minimizing their stopping time. Fixed-budget procedures receive a predetermined number of observations and minimize final simple regret. Adaptive experiments for policy choice use a related distinction between the informational value of an assignment and the payoff generated during the experiment (Kasy and Sautmann, 2021).

Multi-output and Pareto-identification problems provide related examples in which observations are vector valued and the final decision depends on comparisons among several outputs (Crépon et al., 2024; Kone et al., 2025). The commodity taxation problem differs because the observed outputs are not separate welfare criteria. They form an economic gradient field that must be integrated, with a controlled remainder, to compare one scalar latent objective.

**Continuous-armed optimization.** Hierarchical optimistic optimization and the theory of  $X$ -armed bandits organize a continuous action space into a tree of increasingly fine regions and sample regions that remain plausibly optimal (Bubeck et al., 2011b). Bregman Contrast Pruning uses the same broad principle of optimism and hierarchical elimination. Its feedback model is different: welfare is not observed at a queried tax. A welfare confidence interval must first be constructed from ordinary and weighted demand along a directed recovery path.

Dynamic pricing also learns from demand observed at experimentally chosen prices (Besbes and Zeevi, 2009). Revenue at a price is a pointwise function of demand at that same price. Welfare is nonpointwise: private surplus depends on an integral of demand, and under heterogeneous weights the relevant integral is formed from weighted demand. This distinction explains why observations at counterfactual taxes and the geometry of paths between taxes enter the present design.

### 1.3 Outline

Section 2 introduces the quasilinear commodity-tax economy, the ordinary and weighted population fields, and the welfare-contrast identity. Section 3 develops the Bregman geometry of welfare estimation, including quadratic Mahalanobis geometry, rank-deficient curvature, and nonquadratic directional curvature. Section 4 specifies the household purchase observation model, the model class (including a hard family used later for a lower bound), and the conditional covariance of the observed demand vectors. Section 5 presents Bregman Contrast Pruning, develops covariance-adaptive confidence bounds, proves safe pruning, derives fixed-confidence, fixed-budget, localized upper bounds on sample complexity, and tight minimax upper and lower bounds on simple regret. Section 6 discusses the interpretation, limitations, and extensions of the results.

## 2 Economic Setting

This section introduces the economy and the population objects entering the learning problem, including notably, the structure of social welfare with respect to prices. I begin with establishing that under the defined economy, a policymaker chooses a vector of commodity taxes, households respond through their purchases, and social welfare becomes a combination of tax revenue and weighted consumer surplus. I then show that this setting allows social welfare to be recovered from observed demand.

### 2.1 Commodity taxation and household choice

There are  $J \geq 1$  commodities. The normalized feasible bundle set  $\mathcal{Y} \subseteq [0, 1]^J$  is compact, the vector of pretax prices is  $p \in \mathbb{R}_+^J$ , and the feasible tax domain  $\mathcal{X} \subseteq [0, 1]^J$  is compact, convex, and contains 0. Under taxation  $x \in \mathcal{X}$ , households face prices  $p + x$ . Taxes are assumed to be fully passed through to consumers (equivalent to sales taxes).

Let  $(\Theta, \mathcal{F}, \Pi)$  be a probability space of household types. A type  $\theta \in \Theta$  has preferences represented by *quasilinear utility*, i.e.  $U_\theta(y, z) = u_\theta(y) + z$ , where  $u_\theta(y)$  is finite, concave, and upper semicontinuous on  $\mathcal{Y}$ ,  $y \in \mathcal{Y}$  is a commodity bundle, and  $z$  is an untaxed numeraire good (e.x. money). The function  $u_\theta$  is assumed to be finite, concave, and upper semicontinuous on  $\mathcal{Y}$ . Quasilinearity makes the marginal utility of the numeraire constant and removes income effects from the commodity choice. The outside option is represented by  $y = 0$ .

Fix a bounded open convex neighborhood  $\mathcal{O} \supseteq \mathcal{X}$  and an open convex set  $\mathcal{O}^+$  satisfying  $\overline{\mathcal{O}} \subset \mathcal{O}^+$ . Although feasible tax policies remain restricted to  $\mathcal{X}$ , at taxation  $x \in \mathcal{O}$ , type  $\theta$  chooses a bundle  $g_\theta(x)$  and has normalized indirect surplus  $v_\theta(x)$  given by

$$\begin{aligned} g_\theta(x) &\in \arg \max_{y \in \mathcal{Y}} \{u_\theta(y) - \langle p + x, y \rangle\}, \\ v_\theta(x) &= \sup_{y \in \mathcal{Y}} \{u_\theta(y) - \langle p + x, y \rangle\} - \sup_{y \in \mathcal{Y}} \{u_\theta(y) - \langle p, y \rangle\}. \end{aligned} \tag{1}$$

Proposition 2.1 relates  $g_\theta$  and  $v_\theta$  via the envelope theorem (Milgrom and Segal, 2002). To set up this proposition, note that the compactness of  $\mathcal{Y}$  and upper semicontinuity of  $u_\theta$  ensure that the maxima exist. The second term normalizes the zero-tax policy, so  $v_\theta(0) = 0$ . A measurable demand selection is assumed to exist, and the maximizing bundle is unique every type at the tax vectors under consideration almost surely.

**Proposition 2.1** (Envelope representation). *For every  $\theta$ , the function  $v_\theta$  is convex on  $\mathcal{O}$ , and  $-g_\theta(x) \in \partial v_\theta(x)$ . If  $v_\theta$  is differentiable at  $x$ , then*

$$g_\theta(x) = -\nabla v_\theta(x).$$

*Proof.* For fixed  $y \in \mathcal{Y}$ , the map  $x \mapsto u_\theta(y) - \langle p + x, y \rangle$  is affine. Its pointwise supremum is therefore convex, and subtracting the zero-tax value does not affect convexity. If  $g_\theta(x)$  is optimal at  $x$ , evaluating the same bundle at  $z \in \mathcal{O}$  gives  $v_\theta(z) \geq v_\theta(x) - \langle g_\theta(x), z - x \rangle$ . Thus  $-g_\theta(x)$  is a subgradient of  $v_\theta$  at  $x$ . At a differentiability point, the subdifferential is the singleton  $\{\nabla v_\theta(x)\}$ .  $\square$

Demand is deterministic conditional on the full household type, but the type and utility function are not observed by the policymaker. The identity  $g_\theta(x) = -\nabla v_\theta(x)$  allows for utility to be calculated from observed demand.

## 2.2 A converse exact-choice representation

Proposition 2.1 shows that quasilinear utility maximization generates a convex indirect-surplus potential  $v_\theta(x)$ . The converse is also useful: a smooth convex potential whose negative gradient is feasible can be rationalized by a concave quasilinear utility. This representation ensures that the hard family introduced in Section 4 consists of exact utility-maximizing household choices.

**Lemma 2.1** (Exact-choice utility representation). *Let  $v : \mathcal{O}^+ \rightarrow \mathbb{R}$  be differentiable and convex, with  $v(0) = 0$ , and suppose that  $-\nabla v(x) \in \mathcal{Y}$  for every  $x \in \mathcal{O}$ . Define*

$$\tilde{u}(y) := \inf_{z \in \mathcal{O}} \{v(z) + \langle p + z, y \rangle\}, \quad y \in \mathcal{Y}. \quad (2)$$

*Then  $\tilde{u}$  is finite, concave, and upper semicontinuous. Moreover, for every  $x \in \mathcal{O}$ ,*

$$v(x) = \sup_{y \in \mathcal{Y}} \{\tilde{u}(y) - \langle p + x, y \rangle\}, \quad (3)$$

*and the unique maximizing bundle is*

$$g(x) = -\nabla v(x).$$

*Proof.* Compactness of  $\overline{\mathcal{O}}$  and continuity of  $v$  imply that the infimum in equation (2) is finite. Since  $\tilde{u}$  is an infimum of continuous affine functions of  $y$ , it is concave and upper semicontinuous.

Fix  $x \in \mathcal{O}$  and put  $y_x := -\nabla v(x)$ . Convexity gives, for every  $z \in \overline{\mathcal{O}}$ ,

$$v(z) \geq v(x) + \langle \nabla v(x), z - x \rangle = v(x) - \langle y_x, z - x \rangle.$$

Consequently,  $v(z) + \langle p + z, y_x \rangle \geq v(x) + \langle p + x, y_x \rangle$ . The infimum defining  $\tilde{u}(y_x)$  is therefore attained at  $z = x$ , and

$$\tilde{u}(y_x) - \langle p + x, y_x \rangle = v(x). \quad (4)$$

For arbitrary  $y \in \mathcal{Y}$ , evaluating the infimum in (2) at  $z = x$  gives  $\tilde{u}(y) - \langle p + x, y \rangle \leq v(x)$ . Together with equation (4), this proves (3).

Suppose another bundle  $y$  attains equality. Then  $x$  minimizes the differentiable function  $z \mapsto v(z) + \langle z, y \rangle$  over  $\overline{\mathcal{O}}$ . Since  $x \in \mathcal{O}$ , the first-order condition gives  $\nabla v(x) + y = 0$ . Thus  $y = -\nabla v(x) = y_x$ , proving uniqueness.  $\square$

### 2.3 Social welfare and welfare contrasts

The priority weight function  $\omega : \Theta \rightarrow [a, b]$  satisfies  $0 \leq a \leq b < \infty$ . The weight is a normative input chosen by the policymaker as a prespecified function of observed household characteristics. Using  $\omega(\theta)$ , I define the *weighted population potential*  $P(x) = \mathbb{E}_\Pi[\omega(\theta)v_\theta(x)]$  along with its ordinary counterpart  $V(x) = \mathbb{E}_\Pi[v_\theta(x)]$ . Similarly, I define *mean demand*  $G(x) = \mathbb{E}_\Pi[g_\theta(x)]$  and its weighted version  $D(x) = \mathbb{E}_\Pi[\omega(\theta)g_\theta(x)]$ .

Notably, these population level objects retain an envelope representation.

**Proposition 2.2** (Population envelope representation). *The population potentials  $V$  and  $P$  are convex on  $\mathcal{O}$  and satisfy*

$$V(0) = P(0) = 0.$$

Moreover, for every  $x \in \mathcal{O}$ ,

$$-G(x) \in \partial V(x), \quad -D(x) \in \partial P(x).$$

Consequently, if  $V$  and  $P$  are differentiable at  $x$ , then

$$G(x) = -\nabla V(x), \quad D(x) = -\nabla P(x). \quad (5)$$

*Proof.* Convexity follows because  $V$  and  $P$  are, respectively, ordinary and nonnegative weighted expectations of the convex household potentials  $v_\theta$ . The normalization  $v_\theta(0) = 0$  for every  $\theta$  gives

$$V(0) = \mathbb{E}_\Pi[v_\theta(0)] = 0$$

and

$$P(0) = \mathbb{E}_\Pi[\omega(\theta)v_\theta(0)] = 0.$$

Fix  $x, z \in \mathcal{O}$ . By Proposition 2.1,  $-g_\theta(x) \in \partial v_\theta(x)$ , and therefore

$$v_\theta(z) \geq v_\theta(x) - \langle g_\theta(x), z - x \rangle.$$

Taking expectations gives

$$V(z) \geq V(x) - \langle G(x), z - x \rangle.$$

Hence  $-G(x) \in \partial V(x)$ .

Since  $\omega(\theta) \geq 0$ , multiplying the household subgradient inequality by  $\omega(\theta)$  preserves its direction:

$$\omega(\theta)v_\theta(z) \geq \omega(\theta)v_\theta(x) - \langle \omega(\theta)g_\theta(x), z - x \rangle.$$

Taking expectations now gives

$$P(z) \geq P(x) - \langle D(x), z - x \rangle,$$

so  $-D(x) \in \partial P(x)$ .

If  $V$  and  $P$  are differentiable at  $x$ , their subdifferentials are the singletons  $\{\nabla V(x)\}$  and  $\{\nabla P(x)\}$ , respectively. Therefore,

$$-G(x) = \nabla V(x), \quad -D(x) = \nabla P(x),$$

which proves equation (5). □

Public tax revenue is assigned social value one. This normalizes the marginal social value of public funds; a different known value could be incorporated by rescaling the revenue term. The *social welfare* generated by taxation  $x$  is  $W(x) := \langle x, G(x) \rangle + P(x)$ . Thus, an *optimal taxation* is any  $x^* \in \mathcal{X}$  satisfying

$$x^* \in \arg \max_{x \in \mathcal{X}} W(x), \quad W^* := W(x^*) = \max_{x \in \mathcal{X}} W(x).$$

Continuity of  $W$  and compactness of  $\mathcal{X}$  ensure that the set of maximizers is nonempty.

The bounded taxation and bundle domains give a uniform welfare range. This will be used to convert high-probability guarantees into expected simple-regret guarantees.

**Lemma 2.2** (Uniform welfare range). *For every household type  $\theta$  and taxation  $x \in \mathcal{X}$ ,*

$$-J \leq v_\theta(x) \leq 0.$$

*Consequently,*

$$-bJ \leq P(x) \leq 0, \quad 0 \leq \langle x, G(x) \rangle \leq J, \quad -bJ \leq W(x) \leq J.$$

*In particular, the difference between any two welfare levels is at most  $J(1 + b)$ .*

*Proof.* Since  $x$  and every feasible bundle are coordinatewise nonnegative, maximized surplus under taxation  $x$  cannot exceed maximized surplus under the zero-tax policy. Hence  $v_\theta(x) \leq 0$ .

Let  $y_\theta^0$  be the utility maximizing bundle at zero tax. Evaluating the same bundle under taxation  $x$  gives  $v_\theta(x) \geq -\langle x, y_\theta^0 \rangle \geq -J$ . The bounds on  $P$  follow from  $0 \leq \omega(\theta) \leq b$ , while the revenue bounds follow from  $x, G(x) \in [0, 1]^J$ . Adding the two welfare components proves the remaining claims.  $\square$

For  $x, z \in \mathcal{X}$ , the *welfare contrast* is  $W(z) - W(x)$ . A taxation  $x^*$  is optimal if and only if  $W(z) - W(x^*) \leq 0$  for every  $z \in \mathcal{X}$ .

**Proposition 2.3** (Welfare contrast identity). *For every  $x, z \in \mathcal{X}$ ,*

$$\begin{aligned} W(z) - W(x) &= \langle z, G(z) \rangle - \langle x, G(x) \rangle \\ &\quad - \int_0^1 \langle D(x + s(z - x)), z - x \rangle ds. \end{aligned} \tag{6}$$

*Proof.* By the fundamental theorem of calculus and (5),  $P(z) - P(x) = -\int_0^1 \langle D(x + s(z - x)), z - x \rangle ds$ . Adding the difference in tax revenue proves the result.  $\square$

Equation (6) is exact, but it requires weighted mean demand to be known along a continuum of tax vectors. A finite experiment must replace the integral by finitely many first-order increments. The next section shows that the resulting extrapolation error is measured naturally by Bregman divergence.

### 3 Geometry

The contrast identity in Proposition 2.3 requires weighted demand along an entire line segment. A first-order approximation uses only demand observed at one tax. The price of this reduction is a deterministic extrapolation error, and convexity identifies that error as a Bregman divergence.

For a differentiable convex function  $F : \mathcal{O} \rightarrow \mathbb{R}$ , the *Bregman divergence* generated by  $F$  from  $x$  to  $z$  is  $B_F(z, x) := F(z) - F(x) - \langle \nabla F(x), z - x \rangle$ . Its *symmetrized Bregman divergence* is  $S_F(x, z) := B_F(z, x) + B_F(x, z)$ .

Applying the definition to  $P$  and using  $D = -\nabla P$  gives

$$P(z) - P(x) = -\langle D(x), z - x \rangle + B_P(z, x). \quad (7)$$

**Theorem 3.1** (Relative Bregman inequality). *For every  $x, z \in \mathcal{O}$  and  $\omega(\theta) \in [a, b]$ ,*

$$aB_V(z, x) \leq B_P(z, x) \leq bB_V(z, x).$$

*Proof.* For each household type, define  $R_\theta(z, x) := v_\theta(z) - v_\theta(x) + \langle g_\theta(x), z - x \rangle$ . Proposition 2.1 gives  $R_\theta(z, x) \geq 0$ , since  $-g_\theta(x)$  is a subgradient of  $v_\theta$  at  $x$ .

Using  $G = -\nabla V$  and  $D = -\nabla P$  gives  $B_V(z, x) = \mathbb{E}_\Pi[R_\theta(z, x)]$  and  $B_P(z, x) = \mathbb{E}_\Pi[\omega(\theta)R_\theta(z, x)]$ . The result follows by multiplying  $a \leq \omega(\theta) \leq b$  by the nonnegative remainder and taking expectations.  $\square$

Now, assume,  $V, P \in C^2(\mathcal{O})$ , and  $\sup_{x \in \mathcal{O}} \|\nabla^2 V(x)\|_{\text{op}} \leq L_V$ . Expanding the two directed ordinary divergences gives

$$S_V(x, z) = \langle G(x) - G(z), z - x \rangle.$$

In particular,  $B_V(z, x) \leq S_V(x, z)$ . Smoothness also gives

$$B_V(z, x) \leq \frac{L_V}{2} \|z - x\|_2^2. \quad (8)$$

Thus, Equation 8 gives a straightforward bound on the extrapolation error of estimating social welfare at taxation  $z$  from data collected under taxation  $x$ . This bound can be made more explicit after specifying some structure on  $V$ .

#### 3.1 Quadratic population potentials

The population potential is *quadratic* when

$$V(x) = c - d^\top x + \frac{1}{2} x^\top H_V x, \quad G(x) = d - H_V x, \quad (9)$$

where  $H_V$  is the Hessian of  $V$  with respect to  $x$ , and as such is symmetric and positive semi-definite. The normalization  $V(0) = 0$  sets  $c = 0$ , while  $d = G(0)$ . Since  $H_V = \nabla^2 V = -\nabla G$ , it is also known as the *substitution-response matrix*, or the change in demand with respect to change in price.

**Proposition 3.1** (Quadratic Bregman geometry). *If  $V$  has the form in equation (9), then*

$$B_V(z, x) = \frac{1}{2}(z - x)^\top H_V(z - x).$$

*Proof.* Substituting equation (9) and  $\nabla V(x) = H_V x - d$  into the definition of  $B_V$  makes the affine terms cancel.  $\square$

If  $H_V e_j = \lambda_j e_j$ , then a taxation change  $z - x = t e_j$  has cost

$$B_V(z, x) = \frac{1}{2} \lambda_j t^2.$$

Large eigenvalues identify directions in which demand changes sharply and first-order extrapolation is expensive. If  $H \succ 0$ , the change of variables  $y = H_V^{1/2} x$  turns the Bregman geometry into squared Euclidean geometry. If  $H$  is singular, its null space consists of directions with no extrapolation error.

**Proposition 3.2** (Relative Hessian bounds). *Suppose  $V$  is quadratic and  $P \in C^2(\mathcal{O})$ . Then*

$$a H_V \preceq \nabla^2 P(x) \preceq b H_V \quad \text{for every } x \in \mathcal{O}. \quad (10)$$

*Proof.* Fix  $x \in \mathcal{O}$  and  $q \in \mathbb{R}^J$ . Since  $\mathcal{O}$  is open,  $x + tq \in \mathcal{O}$  for all sufficiently small  $|t|$ . Apply Theorem 3.1 with  $z = x + tq$ , divide by  $t^2/2$ , and let  $t \rightarrow 0$ . The resulting quadratic-form inequalities for every  $q$  are equivalent to equation (10).  $\square$

Now,  $P$  can be factorized into a possibly nonlinear part and an affine part. Thus, if  $H_V$  is rank-deficient, a lower dimensional learning problem can be induced.

**Theorem 3.2** (Factorization under rank-deficient curvature). *Let  $r = \text{rank}(H_V)$  and choose a full-row-rank matrix  $A \in \mathbb{R}^{r \times J}$  satisfying  $H = A^\top A$ . There are a function  $\tilde{P} : A\mathcal{O} \rightarrow \mathbb{R}$ , a vector  $q \in \ker A$ , and a scalar  $c_0$  such that*

$$P(x) = \tilde{P}(Ax) + q^\top x + c_0, \quad x \in \mathcal{O}.$$

*Proof.* Let  $h \in \ker A = \ker H$ . Proposition 3.2 gives  $h^\top \nabla^2 P(x) h = 0$ . Since  $\nabla^2 P(x)$  is positive semidefinite, this implies  $\nabla^2 P(x) h = 0$ . For every direction  $r \in \mathbb{R}^J$ ,  $D_r \{h^\top \nabla P(x)\} = r^\top \nabla^2 P(x) h = 0$ . Hence  $x \mapsto h^\top \nabla P(x)$  is constant on the connected set  $\mathcal{O}$ .

Fix  $x_0 \in \mathcal{O}$  and let  $q$  be the orthogonal projection of  $\nabla P(x_0)$  onto  $\ker A$ . Then  $h^\top (\nabla P(x) - q) = 0$  for all  $h \in \ker A$ . If  $Ax = Az$ , then  $z - x \in \ker A$ . Integrating the directional derivative of  $P(\cdot) - q^\top(\cdot)$  along the segment from  $x$  to  $z$  shows that  $P(x) - q^\top x = P(z) - q^\top z$ . This common value defines  $\tilde{P}$  on each fiber of  $A$ .  $\square$

The theorem separates an  $r$ -dimensional nonlinear component, represented by  $\tilde{P}(Ax)$ , from an affine component on the  $(J - r)$ -dimensional null space  $\ker A$ . This reduction is especially relevant when household behavior responds primarily to a small number of composite price indices rather than to each commodity price independently (e.x. responding to a milk price index instead of whole and skim milk prices separately). Low-rank

curvature is not equivalent to a strong-substitutes property, but they offer similar implications in multigood demand. Related query and equilibrium problems under strong substitutes are studied by Milgrom and Strulovici (2009), Lock et al. (2022), Baldwin et al. (2023), and Baldwin et al. (2024).

### 3.2 Homogeneous welfare weights

Suppose  $\omega(\theta) = \bar{\omega}$  almost surely. Then  $P = \bar{\omega}V$  and  $D = \bar{\omega}G$ .

**Proposition 3.3** (Homogeneous weights). *If  $V$  is quadratic, then*

$$\begin{aligned} W(x) &= (1 - \bar{\omega})d^\top x - \frac{2 - \bar{\omega}}{2}x^\top Hx, \\ \nabla W(x) &= (2 - \bar{\omega})G(x) - G(0), \\ W(z) - W(x) &= \langle \nabla W(x), z - x \rangle - (2 - \bar{\omega})B_V(z, x). \end{aligned}$$

*Proof.* Substitute  $P = \bar{\omega}V$  and equation (9) into the welfare definition. Differentiation gives the second identity, and direct expansion gives the third.  $\square$

When  $\bar{\omega} < 2$  and  $H_V \succ 0$ , welfare is strongly concave. If the maximizer is an interior stationary point, then

$$W(x^*) - W(x) = (2 - \bar{\omega})B_V(x, x^*).$$

At a constrained boundary maximizer, strong concavity still gives quadratic growth, but the exact difference also contains the first-order optimality term.

### 3.3 Nonquadratic population potentials

Lifting the quadratic restriction on  $V$  makes the curvature of demand much harder to learn. Under quadratic  $V$ , a number of equivalences lead to useful results.

**Proposition 3.4** (Characterization of quadratic potentials). *The following are equivalent on the connected open set  $\mathcal{O}$ :  $V$  is quadratic,  $G$  is affine, and  $H_V$  is constant.*

*Proof.* Each implication follows by differentiation or integration of a constant Hessian.  $\square$

These equivalences do not generally hold under a nonquadratic  $V$ . In fact, without quadratic  $V$ , a much more general form is needed for  $B_V(z, x)$ . Noticeably, the Hessian  $H_V(x)$  is no longer constant, instead depending on  $x$ .

**Proposition 3.5** (Integral representation). *Let  $h = z - x$ . Then*

$$\begin{aligned} B_V(z, x) &= \int_0^1 (1 - t)h^\top H_V(x + th)h dt, \\ B_V(x, z) &= \int_0^1 t h^\top H_V(x + th)h dt. \end{aligned} \tag{11}$$

*Proof.* Apply Taylor's theorem with integral remainder to  $t \mapsto V(x + th)$  and then to the reversed segment.  $\square$

The Bregman divergence along a continuous path can be written as below.

**Proposition 3.6** (Local Bregman expansion). *For every  $x \in \mathcal{O}$ ,*

$$B_V(x + h, x) = \frac{1}{2} h^\top H_V(x) h + o(\|h\|_2^2)$$

as  $h \rightarrow 0$  with  $x + h \in \mathcal{O}$ .

*Proof.* Continuity gives  $H_V(x + th) = H_V(x) + o(1)$  uniformly in  $t \in [0, 1]$ . Substitute this relation into equation (11).  $\square$

Thus, the bound on the extrapolation error in the nonquadratic case depends on the taxation  $x \in \mathcal{X}$  at which households were sampled.

**Proposition 3.7** (Comparison with a fixed Mahalanobis geometry). *Suppose there are  $Q \succ 0$  and constants  $0 < c_- \leq c_+ < \infty$  such that  $c_- Q \preceq H_V(y) \preceq c_+ Q$  for every  $y \in \mathcal{O}$ . Then, for every  $x, z \in \mathcal{O}$ ,*

$$\frac{c_-}{2} \|z - x\|_Q^2 \leq B_V(z, x) \leq \frac{c_+}{2} \|z - x\|_Q^2, \quad \|q\|_Q^2 := q^\top Q q.$$

*Proof.* Apply the Hessian inequalities pointwise in equation (11).  $\square$

### 3.4 Directional curvature and nonquadraticity

As previously noted, under nonquadratic  $V$ , the Hessian  $H_V(x)$  depends on  $x$ . This induces a path dependency in the extrapolation error for estimating social welfare at a unseen taxation  $z$ .

For  $r \geq 0$ , define the *Hessian modulus*

$$\varpi_V(r) := \sup \left\{ \|\nabla^2 V(x) - \nabla^2 V(z)\|_{\text{op}} : x, z \in \mathcal{X}, \|x - z\|_2 \leq r \right\}.$$

**Proposition 3.8** (Directed and symmetrized Bregman cost). *For  $x, z \in \mathcal{X}$ ,*

$$\left| B_V(z, x) - \frac{1}{2} S_V(x, z) \right| \leq \frac{1}{4} \varpi_V(\|z - x\|_2) \|z - x\|_2^2.$$

*If the Hessian  $H_V$  is  $L_H$ -Lipschitz, then*

$$\left| B_V(z, x) - \frac{1}{2} S_V(x, z) \right| \leq \frac{L_H}{12} \|z - x\|_2^3.$$

*Proof.* Put  $h = z - x$  and  $q(t) = h^\top \nabla^2 V(x + th) h$ . By Proposition 3.5,  $B_V(z, x) - S_V(x, z)/2 = \int_0^1 (1/2 - t) q(t) dt$ . The integral of  $1/2 - t$  is zero, so  $q(1/2)$  may be subtracted inside the integrand. The definition of  $\varpi_V$  and  $\int_0^1 |1/2 - t| dt = 1/4$  give the first bound. Under an  $L_H$ -Lipschitz Hessian, use  $|q(t) - q(1/2)| \leq L_H \|h\|_2^3 |t - 1/2|$  and  $\int_0^1 (t - 1/2)^2 dt = 1/12$ .  $\square$

Thus, the extrapolation error needs to be considered an error along an entire sequence of candidate taxations.

For an integer  $K \geq 1$ , let  $u_j = x + j(z - x)/K$ .

For  $x, z \in \mathcal{X}$ , the continuous line segment joining  $x$  to  $z$  is parameterized by

$$q_{x,z}(t) := x + t(z - x), \quad t \in [0, 1].$$

Because  $\mathcal{X}$  is convex,  $q_{x,z}(t) \in \mathcal{X}$  for every  $t \in [0, 1]$ .

For an integer  $K \geq 1$ , the *equally spaced straight path* from  $x$  to  $z$  with  $K$  directed edges is the sequence

$$\pi_K(x, z) := (u_0, \dots, u_K), \quad u_j := q_{x,z}\left(\frac{j}{K}\right) = x + \frac{j}{K}(z - x), \quad 0 \leq j \leq K.$$

Thus  $u_0 = x$ ,  $u_K = z$ , and every edge has the same displacement and Euclidean length:

$$u_{j+1} - u_j = \frac{z - x}{K}, \quad \|u_{j+1} - u_j\|_2 = \frac{\|z - x\|_2}{K}.$$

The path is directed from  $x$  to  $z$ ; its reversal is  $\pi_K(z, x)$ . Under a nonquadratic potential, these two directions may have different cumulative Bregman costs.

## 4 Statistical Model

The economic setting describes how a household type responds to a taxation. I now specify how those responses are observed in an experiment. Specifically, I define the structural model class formally, and construct the hard exact-choice family within that model class that will later be used to construct a lower bound on minimax expected regret of the BCP algorithm.

### 4.1 Household model classes

Fix the domains  $\mathcal{X}, \mathcal{Y}$ , the pretax price vector  $p$ , the common open neighborhood  $\mathcal{O}$ , and the constants  $a, b, L_V$ . A *household environment* is a tuple

$$M = (\Theta_M, \mathcal{F}_M, \Pi_M, u_M, \omega_M),$$

where  $u_M : \Theta_M \times \mathcal{Y} \rightarrow \mathbb{R}$  denotes the direct-utility mapping. The tuple is *admissible* when the following conditions hold.

- (M1)  $(\Theta_M, \mathcal{F}_M, \Pi_M)$  is a probability space.
- (M2) The map  $u_M$  is  $\mathcal{F}_M \otimes \mathcal{B}(\mathcal{Y})$ -measurable, and  $u_{\theta, M}(\cdot) := u_M(\theta, \cdot)$  is finite, concave, and upper semicontinuous for every  $\theta$ .
- (M3) For every  $x \in \mathcal{O}$ , the utility-maximization problem in (1) admits a measurable selected demand  $g_{\theta, M}(x) \in \mathcal{Y}$ , unique for  $\Pi_M$ -almost every type. The corresponding normalized indirect surplus is denoted by  $v_{\theta, M}(x)$ .
- (M4) The priority weight function  $\omega_M : \Theta_M \rightarrow [a, b]$  is measurable and is a prespecified function of observed household characteristics.
- (M5) The population fields induced by  $M$  are finite on  $\mathcal{O}$ ,  $V_M, P_M \in C^2(\mathcal{O})$ ,

$$G_M = -\nabla V_M, \quad D_M = -\nabla P_M, \quad \sup_{x \in \mathcal{O}} \|\nabla^2 V_M(x)\|_{\text{op}} \leq L_V.$$

The *household model class* is the set

$$\mathcal{M} := \mathcal{M}(\mathcal{X}, \mathcal{Y}, \mathcal{O}; p, a, b, L_V) := \{M : M \text{ satisfies (M1)–(M5)}\}.$$

For  $M \in \mathcal{M}$ , write  $\mathbb{P}_M$  and  $\mathbb{E}_M$  for probability and expectation under the complete sequential experiment. The class is nonparametric:  $M$  records the household type distribution and the entire collection of utilities determining behavior jointly at every counterfactual taxation.

### 4.2 Exact-choice household purchase observations

A household environment  $M$  induces household purchase behaviors on each type  $\theta \in \Theta$ , allowing for the definition of laws on the samples of household purchases collected in each round of a sequential experiment.

For a fixed taxation  $x \in \mathcal{X}$ , the experiment draws a type  $\theta \sim \Pi_M$  and observes

$$(Y, \omega) = (g_{\theta, M}(x), \omega_M(\theta)).$$

Thus a household's purchase behavior is the household's exact utility-maximizing bundle conditional on its full type.

Let  $Q_{M,1}^x$  denote the law of one household purchase at taxation  $x$ . Equivalently, for every Borel set  $B \subseteq \mathcal{Y} \times [a, b]$ ,

$$Q_{M,1}^x(B) = \Pi_M \{ \theta : (g_{\theta, M}(x), \omega_M(\theta)) \in B \}.$$

If  $(Y, \omega) \sim Q_{M,1}^x$ , then  $\mathbb{E}_M[Y] = G_M(x)$  and  $\mathbb{E}_M[\omega Y] = D_M(x)$ .

Sampling rounds are indexed by  $t \geq 1$ . Let  $\mathcal{R}$  contain any internal randomization used by the experiment. The information available after round  $t$  is

$$\mathcal{F}_t := \sigma(\mathcal{R}, X_s, (Y_{si}, \omega_{si})_{i=1}^m : 1 \leq s \leq t), \quad t \geq 0.$$

The taxation  $X_t \in \mathcal{X}$  is  $\mathcal{F}_{t-1}$ -measurable. Conditional on  $\mathcal{F}_{t-1}$ , round  $t$  contains  $m$  fresh households,

$$\theta_{t1}, \dots, \theta_{tm} \stackrel{\text{i.i.d.}}{\sim} \Pi_M,$$

(conditionally) independent of past samples. Their observations are  $Y_{ti} = g_{\theta_{ti}, M}(X_t)$  and  $\omega_{ti} = \omega_M(\theta_{ti})$ .

The priority weight is a prespecified function of household characteristics observed by the policymaker. It is therefore a normative input rather than a latent behavioral outcome, and the weighted purchase  $\omega_{ti} Y_{ti}$  is observable.

At a fixed taxation  $x$ , the  $m$  household observations in a round  $t$  are independent and identically distributed under  $Q_{M,1}^x$ . Hence the complete-wave law is

$$Q_{M,m}^x = (Q_{M,1}^x)^{\otimes m}.$$

Relative entropy is additive across independent product distributions.

**Lemma 4.1** (Relative entropy of one experimental wave). *For every fixed taxation  $x \in \mathcal{X}$  and every pair of environments  $M_0, M_1 \in \mathcal{M}$ ,*

$$\text{KL}(Q_{M_0,m}^x \parallel Q_{M_1,m}^x) = m \text{KL}(Q_{M_0,1}^x \parallel Q_{M_1,1}^x). \quad (12)$$

*The equality is understood in the extended real numbers.*

*Proof.* Write

$$Q_0 := Q_{M_0,1}^x, \quad Q_1 := Q_{M_1,1}^x.$$

First suppose that  $Q_0 \ll Q_1$ , and let

$$L(o) := \frac{dQ_0}{dQ_1}(o)$$

be the Radon–Nikodym derivative of the one-household law under  $M_0$  with respect to that under  $M_1$ . By the product structure,

$$\frac{dQ_0^{\otimes m}}{dQ_1^{\otimes m}}(o_1, \dots, o_m) = \prod_{i=1}^m L(o_i).$$

Therefore,

$$\begin{aligned} \text{KL}(Q_0^{\otimes m} \parallel Q_1^{\otimes m}) &= \int \log \left\{ \prod_{i=1}^m L(o_i) \right\} Q_0^{\otimes m}(d(o_1, \dots, o_m)) \\ &= \sum_{i=1}^m \int \log L(o_i) Q_0^{\otimes m}(d(o_1, \dots, o_m)) \\ &= \sum_{i=1}^m \int \log L(o) Q_0(do) \\ &= m \text{KL}(Q_0 \parallel Q_1). \end{aligned}$$

If  $Q_0 \not\ll Q_1$ , there is a measurable set  $A$  such that  $Q_1(A) = 0$  and  $Q_0(A) > 0$ . Then

$$Q_1^{\otimes m}(A \times \mathcal{Z}^{m-1}) = 0, \quad Q_0^{\otimes m}(A \times \mathcal{Z}^{m-1}) > 0,$$

where  $\mathcal{Z} := \mathcal{Y} \times [a, b]$  is the one-household observation space. Thus  $Q_0^{\otimes m} \not\ll Q_1^{\otimes m}$ , and both sides of equation (12) are infinite.  $\square$

Consequently, for any two environments,

$$\text{KL}(Q_{M_0, m}^x \parallel Q_{M_1, m}^x) = m \text{KL}(Q_{M_0, 1}^x \parallel Q_{M_1, 1}^x).$$

The *sample mean demand* and *sample mean weighted demand* in wave  $t$  are

$$\hat{G}_t := \frac{1}{m} \sum_{i=1}^m Y_{ti}, \quad \hat{D}_t := \frac{1}{m} \sum_{i=1}^m \omega_{ti} Y_{ti}.$$

They satisfy  $\mathbb{E}_M[\hat{G}_t \mid \mathcal{F}_{t-1}, X_t] = G_M(X_t)$  and  $\mathbb{E}_M[\hat{D}_t \mid \mathcal{F}_{t-1}, X_t] = D_M(X_t)$ .

### 4.3 Conditional covariance and bounded sampling noise

Write the joint household observation and its mean, conditional on  $\mathcal{F}_{t-1}$ , as

$$Z_{ti} := \begin{pmatrix} Y_{ti} \\ \omega_{ti} Y_{ti} \end{pmatrix} \in \mathbb{R}^{2J}, \quad \mu_M(x) := \begin{pmatrix} G_M(x) \\ D_M(x) \end{pmatrix}.$$

The *conditional household covariance* at taxation  $x$  is

$$\Sigma_M(x) := \text{Cov}_M(Z_{ti} \mid X_t = x) \in \mathbb{S}_+^{2J}.$$

Set

$$\sigma^2 := J(1 + b^2).$$

Since  $Y_{ti} \in [0, 1]^J$  and  $0 \leq \omega_{ti} \leq b$ ,  $\|Z_{ti}\|_2 \leq \sigma$  almost surely. For every  $\xi \in \mathbb{R}^{2J}$ ,  $\xi^\top \Sigma_M(x) \xi = \text{Var}_M(\xi^\top Z_{ti} \mid X_t = x) \leq \sigma^2 \|\xi\|_2^2$ . Hence

$$0 \preceq \Sigma_M(x) \preceq \sigma^2 I_{2J}.$$

The scalar proxy  $\sigma^2$  is therefore a valid isotropic upper bound, but a particular welfare comparison depends on the sharper directional variance  $\xi^\top \Sigma_M(x) \xi$ .

**Proposition 4.1** (Sub-Gaussian household observations). *For every  $\xi \in \mathbb{R}^{2J}$  and  $s \in \mathbb{R}$ ,*

$$\mathbb{E}_M [\exp \{s \langle \xi, Z_{ti} - \mu_M(X_t) \rangle\} \mid \mathcal{F}_{t-1}] \leq \exp \left\{ \frac{s^2 \sigma^2 \|\xi\|_2^2}{2} \right\}. \quad (13)$$

*Proof.* Conditional on  $\mathcal{F}_{t-1}$ , the scalar projection  $\langle \xi, Z_{ti} \rangle$  lies in an interval of length at most  $2\sigma \|\xi\|_2$ . Hoeffding's lemma (Boucheron et al., 2013) gives (13).  $\square$

If a fixed taxation  $x$  is used for  $n$  independent waves and  $\bar{Z}_n(x)$  is the average of the resulting  $mn$  household observations, conditional independence gives

$$\mathbb{E}_M [\exp \{s \langle \xi, \bar{Z}_n(x) - \mu_M(x) \rangle\}] \leq \exp \left\{ \frac{s^2 \sigma^2 \|\xi\|_2^2}{2mn} \right\}.$$

The cohort size  $m$  and the number of repeated waves  $n$  enter through their product.

## 4.4 A structural hard family

This subsection constructs the economic environments used in the lower bound on expected simple regret later in Section 5. I specify the population potentials, the distribution of household types, and the exact choices made by every type. This subsection simply introduces the hard family induced by such an economic environment. The family's properties are analyzed in Section 5.

The hard family is defined to be within a specific model class  $\mathcal{M}_{\text{ex}}$ . Fix  $\rho > 0$ , take  $\mathcal{O} = (-\rho, 1 + \rho)^J$  and  $\mathcal{O}^+ = (-2\rho, 1 + 2\rho)^J$ , and define

$$\mathcal{M}_{\text{ex}} := \mathcal{M}([0, 1]^J, [0, 1]^J, (-\rho, 1 + \rho)^J; p, a, b, L_V),$$

where  $L_V > 0$  and  $[a, b] \cap (0, 1) \neq \emptyset$ .

### 4.4.1 Baseline and localized welfare alternatives

Choose  $\bar{\omega} \in [a, b] \cap (0, 1)$  and fix  $r_0 \in (0, 1/4)$ . On the interval  $(-2\rho, 1 + 2\rho)$ , define

$$f(t) := \begin{cases} -t^{\bar{\omega}}, & t \geq r_0, \\ -r_0^{\bar{\omega}} - \bar{\omega} r_0^{\bar{\omega}-1}(t - r_0) + \frac{\bar{\omega}(1 - \bar{\omega})}{2} r_0^{\bar{\omega}-2}(t - r_0)^2, & t < r_0. \end{cases}$$

The value, first derivative, and second derivative agree at  $r_0$ . Thus  $f$  is twice continuously differentiable. It is decreasing and strictly convex. Direct calculation gives

$$-tf'(t) + \bar{\omega}f(t) = \begin{cases} 0, & t \geq r_0, \\ -\frac{\bar{\omega}(1 - \bar{\omega})(2 - \bar{\omega})}{2} r_0^{\bar{\omega}-2}(r_0 - t)^2, & t < r_0. \end{cases} \quad (14)$$

Choose  $c > 0$  sufficiently small and define

$$V_0(x) := c \sum_{j=1}^J \{f(x_j) - f(0)\}, \quad P_0(x) := \bar{\omega} V_0(x). \quad (15)$$

The baseline mean demand is  $G_0 = -\nabla V_0$ . Since  $-f'$  is continuous and strictly positive on  $[0, 1]$ ,  $c$  may be chosen so that there is  $\nu \in (0, 1/8)$  satisfying

$$4\nu \leq G_{0,j}(x) \leq 1 - 4\nu \quad \text{for every } j \text{ and } x \in \mathcal{O}. \quad (16)$$

The same  $c$  may be chosen to ensure

$$\sup_{x \in \mathcal{O}} \|\nabla^2 V_0(x)\|_{\text{op}} \leq \frac{L_V}{2}.$$

Because the welfare weight is homogeneous, baseline welfare is  $W_0(x) = -x^\top \nabla V_0(x) + \bar{\omega} V_0(x)$ . Equations (15) and (14) imply

$$W_0(x) \leq w_0 := -cJ\bar{\omega}f(0) \quad \text{for every } x \in [0, 1]^J, \quad (17)$$

with equality whenever every coordinate of  $x$  is at least  $r_0$ .

Let

$$\mathcal{I} := [1/3, 2/3]^J.$$

For every sufficiently small  $h > 0$ , the interior cube can be packed with pairwise disjoint cubes

$$R_k := z_k + h[-1/2, 1/2]^J \subset \mathcal{I}, \quad 1 \leq k \leq N_h, \quad N_h \geq c_0 h^{-J},$$

where  $c_0 > 0$  depends only on  $J$ .

Choose  $\psi \in C_c^\infty((-1/2, 1/2)^J)$  satisfying  $\psi(0) = 0$  and  $\nabla \psi(0) = -e_1$ . Set

$$K_1 := \sup_s \|\nabla \psi(s)\|_2, \quad K_2 := \sup_s \|\nabla^2 \psi(s)\|_{\text{op}}.$$

Fix a mixture probability  $\chi \in (0, 1)$ . Since  $V_0$  is strictly convex and  $\mathcal{I}$  is compact, there is  $\kappa_0 > 0$  such that

$$\nabla^2 V_0(x) \succeq \kappa_0 I_J \quad \text{for every } x \in \mathcal{I}. \quad (18)$$

Choose  $\eta > 0$  such that

$$\frac{\eta K_2}{1 - \chi} \leq \frac{\kappa_0}{2}, \quad \eta K_2 \leq \frac{L_V}{2}. \quad (19)$$

For each packed region, define

$$\phi_{k,h}(x) := \eta h^2 \psi\left(\frac{x - z_k}{h}\right), \quad V_k := V_0 + \phi_{k,h}, \quad P_k := \bar{\omega} V_k.$$

The perturbation is supported in  $R_k$ , and

$$\|\nabla \phi_{k,h}(x)\|_2 \leq \eta h K_1, \quad \|\nabla^2 \phi_{k,h}(x)\|_{\text{op}} \leq \eta K_2.$$

The second condition in (19) ensures that every  $V_k$  satisfies the same  $L_V$ -smoothness bound as the upper-bound class.

The welfare perturbation is

$$W_k(x) - W_0(x) = -x^\top \nabla \phi_{k,h}(x) + \bar{\omega} \phi_{k,h}(x).$$

It vanishes outside  $R_k$ . At the center,

$$W_k(z_k) - W_0(z_k) = \eta h z_{k,1} \geq \frac{\eta h}{3}.$$

Since  $W_0(z_k) = w_0$ , it follows that

$$\max_{x \in R_k} W_k(x) \geq w_0 + c_1 h, \quad c_1 := \frac{\eta}{3}, \quad (20)$$

whereas every taxation outside  $R_k$  has welfare at most  $w_0$ .

#### 4.4.2 Exact-choice heterogeneous households

The hard family uses two latent preference components. The component label is not observed by the policymaker and should not be interpreted as an observed demographic category. Component  $A$  supplies a common density shared by the baseline and all alternatives. Component  $B$  carries the localized perturbation.

Choose symmetric continuously differentiable densities  $f_A$  and  $f_B$  on  $\mathbb{R}^J$ , both having mean zero. Let  $f_B$  be supported in  $[-\nu/4, \nu/4]^J$ . Let  $f_A$  be supported in  $[-\nu, \nu]^J$  and satisfy

$$f_A(\xi) \geq c_A > 0 \quad \text{for every } \xi \in [-\nu/2, \nu/2]^J. \quad (21)$$

A household type is  $\theta = (R, \xi)$ , where  $R \in \{A, B\}$  is an unobserved component label. The population law is

$$\Pi := \chi(\delta_A \otimes F_A) + (1 - \chi)(\delta_B \otimes F_B),$$

where  $F_A$  and  $F_B$  have densities  $f_A$  and  $f_B$ . This type distribution is nonatomic because both conditional taste distributions are nonatomic.

Under the baseline environment, define the individual indirect-surplus potentials by

$$v_{A,\xi,0}(x) = v_{B,\xi,0}(x) := V_0(x) - \langle \xi, x \rangle.$$

Under alternative environment  $M_k$ , component  $A$  is unchanged, while component  $B$  has potential

$$v_{B,\xi,k}(x) := V_0(x) + \frac{\phi_{k,h}(x)}{1 - \chi} - \langle \xi, x \rangle.$$

Every household receives the common observed welfare weight  $\bar{\omega}$ .

Outside  $R_k$ , the perturbation vanishes. Inside  $R_k$ , (18) and (19) give

$$\nabla^2 v_{B,\xi,k}(x) = \nabla^2 V_0(x) + \frac{\nabla^2 \phi_{k,h}(x)}{1 - \chi} \succeq \frac{\kappa_0}{2} I_J.$$

Thus every individual potential in the construction is convex.

The exact demand generated by type  $(R, \xi)$  is

$$g_{R,\xi,k}(x) := -\nabla v_{R,\xi,k}(x).$$

For component  $A$  under every environment and for both components under the baseline,  $g_{R,\xi,0}(x) = G_0(x) + \xi$ . For component  $B$  under alternative  $M_k$ ,

$$g_{B,\xi,k}(x) = G_0(x) + \xi - \frac{\nabla \phi_{k,h}(x)}{1 - \chi}. \quad (22)$$

Reduce the maximal admissible value  $h_0$  so that  $\eta h_0 K_1 / (1 - \chi) \leq \nu/4$ . Then (16), the supports of  $f_A$  and  $f_B$ , and (22) imply

$$g_{R,\xi,k}(x) \in [0, 1]^J \quad (23)$$

for every  $x \in \mathcal{O}$ , every type, and every environment in the construction.

Lemma 2.1 supplies a finite, concave, upper-semicontinuous direct utility  $u_{R,\xi,k}$  whose unique utility-maximizing bundle is  $g_{R,\xi,k}(x)$ . The mapping from  $(R, \xi)$  to this utility and its choice is measurable because the objective in (2) is jointly continuous in  $(\xi, y, z)$  and minimized over a common compact set. Thus the constructed environments satisfy the exact-choice requirements of  $\mathcal{M}_{\text{ex}}$ .

Since both taste distributions have mean zero, the population potential under the baseline is  $V_0$ . Under alternative  $M_k$ ,

$$\begin{aligned} \mathbb{E}_{\Pi}[v_{\theta,k}(x)] &= \chi V_0(x) + (1 - \chi) \left\{ V_0(x) + \frac{\phi_{k,h}(x)}{1 - \chi} \right\} \\ &= V_0(x) + \phi_{k,h}(x) = V_k(x). \end{aligned} \quad (24)$$

Since every household has weight  $\bar{\omega}$ , the weighted population potential is

$$P_k = \bar{\omega} V_k. \quad (25)$$

The structural and welfare properties of the hard family can now be summarized.

**Proposition 4.2** (Exact-choice localized welfare alternatives). *There are constants  $c_0, c_1, h_0 > 0$  such that, for every  $0 < h \leq h_0$ , the class  $\mathcal{M}_{\text{ex}}$  contains a baseline environment  $M_0$  and at least  $N_h \geq c_0 h^{-J}$  alternatives  $M_1, \dots, M_{N_h}$  associated with pairwise disjoint regions  $R_1, \dots, R_{N_h}$  satisfying the following properties.*

1. *Every observed bundle is the household's unique exact utility-maximizing choice under a concave quasilinear utility.*
2. *Every household has welfare weight  $\bar{\omega} \in [a, b] \cap (0, 1)$ .*
3.  *$W_{M_0}(x) \leq w_0$  for every  $x$ , and  $W_{M_0}(x) = w_0$  throughout every  $R_k$ .*
4.  *$W_{M_k} = W_{M_0}$  outside  $R_k$ , while*

$$\max_{x \in R_k} W_{M_k}(x) \geq w_0 + c_1 h.$$

*All environments satisfy the same demand, priority weight, and population-smoothness restrictions defining  $\mathcal{M}_{\text{ex}}$ .*

*Proof.* Exact utility maximization follows from Lemma 2.1 and (23). The population identities are (24) and (25). The welfare statements follow from (17) and (20). The choices of  $c$ ,  $\eta$ , and  $h_0$  give the common smoothness and feasibility bounds.  $\square$

The difficulty of this family comes from the combination of many possible perturbation locations and a weak, localized observational signal. Under alternative  $M_k$ , both household behavior and social welfare agree exactly with the baseline at every taxation outside  $R_k$ . Observations collected outside the perturbed region therefore provide no information about whether  $M_k$  or the baseline environment is generating the data.

Even within  $R_k$ , the alternative changes only the demand of latent component  $B$ , and the resulting displacement is of order  $h$ . Component  $A$  remains unchanged and supplies a common positive-density component to the baseline and alternative observation laws. The two environments consequently generate substantially overlapping purchase distributions despite having different welfare-maximizing regions. Therefore, an experiment must achieve two things simultaneously: locate the relevant region among approximately  $h^{-J}$  possibilities and then detect a small change in household behavior after sampling inside that region.

## 5 Experimental Design

At each experimental round, the policymaker observes household purchases at one taxation but must compare welfare over the continuous domain  $\mathcal{X}$ . The Bregman geometry developed in Section 3 controls deterministic error when welfare information is transported between taxes. The conditional covariance developed in Section 4 controls the number of repeated rounds needed to estimate the corresponding contrasts. This section combines those two ingredients in *Bregman Contrast Pruning*.

For the algorithmic results, specialize to  $\mathcal{X} = [0, 1]^J$  and retain the zero-tax policy as a fixed reference. Since  $P(0) = 0$  and tax revenue vanishes at 0, one has  $W(0) = 0$ . A run of the fixed-confidence algorithm is indexed by a terminal epoch  $L \in \mathbb{N}$  and confidence level  $\delta \in (0, 1)$ .

### 5.1 Recovery graphs and pathwise welfare estimates

A *finite directed graph* is a pair  $\mathcal{G} = (\mathcal{U}, \mathcal{E})$ , where  $\mathcal{U} \subseteq \mathcal{X}$  is a finite set of tax nodes and  $\mathcal{E} \subseteq \mathcal{U} \times \mathcal{U}$  is a finite set of directed edges. A *directed path* is a sequence  $\pi = (u_0, \dots, u_K)$  satisfying  $(u_j, u_{j+1}) \in \mathcal{E}$  for  $0 \leq j < K$ . Direction matters because  $B_V(v, u)$  and  $B_V(u, v)$  need not agree under a nonquadratic population potential.

The first lemma converts tax-space resolution into welfare resolution.

**Lemma 5.1** (Lipschitz continuity of welfare). *The welfare function  $W(x)$  is  $L_W$ -Lipschitz on  $\mathcal{X}$ , where*

$$L_W := \sqrt{J} (1 + b + L_V).$$

*Proof.* Differentiating  $W(x) = x^\top G(x) + P(x)$  gives  $\nabla W(x) = G(x) + \nabla G(x)^\top x - D(x)$ . Since purchases lie in  $[0, 1]^J$ ,  $\|G(x)\|_2 \leq \sqrt{J}$  and  $\|D(x)\|_2 \leq b\sqrt{J}$ . Moreover,  $\|x\|_2 \leq \sqrt{J}$  and  $\|\nabla G(x)\|_{\text{op}} = \|\nabla^2 V(x)\|_{\text{op}} \leq L_V$ . Hence  $\|\nabla W(x)\|_2 \leq L_W$ , and the mean-value inequality proves the result.  $\square$

For  $\ell \geq 0$ , let  $\gamma_\ell := 2^{-\ell}$ . The *dyadic lattice* is

$$\mathcal{Q}_\ell := \{0, \gamma_\ell, 2\gamma_\ell, \dots, 1\}^J.$$

Let  $\mathcal{C}_\ell$  be the corresponding partition of  $[0, 1]^J$  into dyadic cubes of side length  $\gamma_\ell$ . Each cell  $C \in \mathcal{C}_\ell$  has a representative  $x_C \in \mathcal{Q}_\ell$  satisfying

$$\sup_{x \in C} \|x - x_C\|_2 \leq \sqrt{J} \gamma_\ell.$$

The dyadic subcells of  $C$  are denoted by  $\text{Children}(C)$ .

For  $x \in \mathcal{Q}_\ell$ , let  $\pi_\ell(x)$  be the canonical monotone lattice path from 0 to  $x$ : coordinates are increased in a fixed order, one at a time, in increments of  $\gamma_\ell$ . Every edge has Euclidean length  $\gamma_\ell$ , and the path contains at most  $J/\gamma_\ell$  edges.

Let  $\mathcal{C}_\ell^{\text{act}}$  be the cells active at epoch  $\ell$ . Define

$$\begin{aligned}\mathcal{U}_\ell &:= \bigcup_{C \in \mathcal{C}_\ell^{\text{act}}} \text{nodes}(\pi_\ell(x_C)), \\ \mathcal{E}_\ell &:= \bigcup_{C \in \mathcal{C}_\ell^{\text{act}}} \text{edges}(\pi_\ell(x_C)), \\ \mathcal{G}_\ell &:= (\mathcal{U}_\ell, \mathcal{E}_\ell).\end{aligned}$$

A node lying on several paths is sampled once during the epoch, and its observations are reused in every policy comparison whose path contains that node.

Let  $T_\ell$  be the round at which epoch  $\ell$  begins and set

$$\mathcal{H}_\ell := \mathcal{F}_{T_\ell}.$$

The active cells, recovery graph, and paths at epoch  $\ell$  are  $\mathcal{H}_\ell$ -measurable.

For an active cell  $C$  and graph node  $u$ , define the deterministic path coefficient

$$\mathbf{a}_{\ell, C, u} := \begin{pmatrix} \mathbb{1}\{u = x_C\}x_C \\ - \sum_{(r, s) \in \pi_\ell(x_C): r=u} (s - r) \end{pmatrix} \in \mathbb{R}^{2J}.$$

For the canonical path, the sum in the second block contains either one outgoing edge displacement or no term.

## 5.2 Covariance-adaptive sampling

Split the confidence budget as

$$\delta_\Sigma + \delta_\mu = \delta, \quad \delta_\Sigma, \delta_\mu > 0. \quad (26)$$

The lattice contains at most

$$K_\ell^{\text{grid}} := (1 + \gamma_\ell^{-1})^J$$

nodes, and its positive-coordinate nearest-neighbor graph contains at most  $JK_\ell^{\text{grid}}$  directed edges. Define

$$\zeta_\ell := \log \left( \frac{4(L+1)(J+1)K_\ell^{\text{grid}}}{\delta_\mu} \right).$$

The following equal-allocation count will serve as a worst-case fallback and as the scale of the covariance pilot:

$$n_\ell^{\text{iso}} := \left\lceil \frac{16\sigma^2 J \zeta_\ell}{m\gamma_\ell^2} + \frac{4\sigma\sqrt{J}\zeta_\ell}{3m\gamma_\ell} \right\rceil. \quad (27)$$

At every epoch node, collect

$$n_\ell^{\text{pil}} := \max \left\{ \left\lceil \frac{2}{m} \right\rceil, \lceil (n_\ell^{\text{iso}})^{2/3} \rceil \right\}$$

independent pilot rounds, and put

$$q_\ell := \left\lfloor \frac{mn_\ell^{\text{pil}}}{2} \right\rfloor.$$

Thus  $q_\ell \geq 1$ . Pair the first  $2q_\ell$  pilot household observations at node  $u$  and define

$$\widehat{\Sigma}_\ell^{\text{pil}}(u) := \frac{1}{2q_\ell} \sum_{r=1}^{q_\ell} \left( Z_{\ell,u,2r-1}^{\text{pil}} - Z_{\ell,u,2r}^{\text{pil}} \right) \left( Z_{\ell,u,2r-1}^{\text{pil}} - Z_{\ell,u,2r}^{\text{pil}} \right)^\top.$$

The paired construction avoids estimating the unknown mean before estimating covariance.

Set

$$\alpha_\ell := \frac{\delta_\Sigma}{(L+1)K_\ell^{\text{grid}}}$$

and

$$\xi_{\Sigma,\ell} := 4\sqrt{2}J\sigma^2 \sqrt{\frac{\log(8J^2/\alpha_\ell)}{q_\ell}}.$$

**Lemma 5.2** (Simultaneous pilot covariance bound). *Conditional on  $\mathcal{H}_\ell$ ,*

$$\mathbb{P}_M \left\{ \Sigma_M(u) \preceq \widehat{\Sigma}_\ell^{\text{pil}}(u) + \xi_{\Sigma,\ell} I_{2J} \text{ for every } u \in \mathcal{U}_\ell \mid \mathcal{H}_\ell \right\} \geq 1 - \frac{\delta_\Sigma}{L+1}. \quad (28)$$

*Proof.* Fix  $u \in \mathcal{U}_\ell$  and condition on  $\mathcal{H}_\ell$ . Since  $u$  and the epoch graph are  $\mathcal{H}_\ell$ -measurable, the pilot observations collected at  $u$  are conditionally independent and identically distributed, with conditional mean  $\mu_M(u)$  and covariance  $\Sigma_M(u)$ .

For one pilot pair, define

$$R_r := \frac{1}{2} \left( Z_{\ell,u,2r-1}^{\text{pil}} - Z_{\ell,u,2r}^{\text{pil}} \right) \left( Z_{\ell,u,2r-1}^{\text{pil}} - Z_{\ell,u,2r}^{\text{pil}} \right)^\top.$$

Then

$$\widehat{\Sigma}_\ell^{\text{pil}}(u) = \frac{1}{q_\ell} \sum_{r=1}^{q_\ell} R_r.$$

Writing the two observations in a pair as  $Z_1$  and  $Z_2$ , conditional independence gives

$$\begin{aligned} \mathbb{E}_M[R_r \mid \mathcal{H}_\ell] &= \frac{1}{2} \mathbb{E}_M \left[ (Z_1 - Z_2)(Z_1 - Z_2)^\top \mid \mathcal{H}_\ell \right] \\ &= \Sigma_M(u). \end{aligned}$$

Since each joint observation has Euclidean norm at most  $\sigma$ , every coordinate of  $Z_1 - Z_2$  has absolute value at most  $2\sigma$ . Consequently, every entry of  $R_r$  lies in  $[-2\sigma^2, 2\sigma^2]$ . Hoeffding's inequality (Boucheron et al., 2013), followed by a union bound over the  $(2J)^2$  matrix entries, implies that, with conditional probability at least  $1 - \alpha_\ell$ ,

$$\max_{j,k} \left| \widehat{\Sigma}_{\ell,jk}^{\text{pil}}(u) - \Sigma_{M,jk}(u) \right| \leq 2\sqrt{2}\sigma^2 \sqrt{\frac{\log(8J^2/\alpha_\ell)}{q_\ell}}.$$

For every  $d \times d$  matrix  $A$ ,

$$\|A\|_{\text{op}} \leq d \max_{j,k} |A_{jk}|.$$

Applying this with  $d = 2J$  gives

$$\left\| \widehat{\Sigma}_\ell^{\text{pil}}(u) - \Sigma_M(u) \right\|_{\text{op}} \leq \xi_{\Sigma,\ell}.$$

Now set

$$E_u := \widehat{\Sigma}_\ell^{\text{pil}}(u) - \Sigma_M(u).$$

The matrix  $E_u$  is symmetric. Hence

$$-\|E_u\|_{\text{op}} I_{2J} \preceq E_u \preceq \|E_u\|_{\text{op}} I_{2J}.$$

On the preceding event,

$$-\xi_{\Sigma,\ell} I_{2J} \preceq \widehat{\Sigma}_\ell^{\text{pil}}(u) - \Sigma_M(u).$$

Rearranging gives

$$\Sigma_M(u) \preceq \widehat{\Sigma}_\ell^{\text{pil}}(u) + \xi_{\Sigma,\ell} I_{2J}.$$

Thus the desired covariance upper bound fails at a fixed node with conditional probability at most  $\alpha_\ell$ . Since  $|\mathcal{U}_\ell| \leq K_\ell^{\text{grid}}$ , a conditional union bound gives

$$\begin{aligned} \mathbb{P}_M \left\{ \exists u \in \mathcal{U}_\ell : \Sigma_M(u) \not\preceq \widehat{\Sigma}_\ell^{\text{pil}}(u) + \xi_{\Sigma,\ell} I_{2J} \mid \mathcal{H}_\ell \right\} \\ \leq K_\ell^{\text{grid}} \alpha_\ell = \frac{\delta_\Sigma}{L+1}. \end{aligned}$$

Taking complements proves equation (28).  $\square$

Define

$$\overline{\Sigma}_\ell(u) := \widehat{\Sigma}_\ell^{\text{pil}}(u) + \xi_{\Sigma,\ell} I_{2J}$$

and, for  $\mathbf{a} \in \mathbb{R}^{2J}$ ,

$$\bar{s}_{\ell,u}^2(\mathbf{a}) := \min \left\{ \mathbf{a}^\top \overline{\Sigma}_\ell(u) \mathbf{a}, \sigma^2 \|\mathbf{a}\|_2^2 \right\}. \quad (29)$$

On the pilot event, both entries of the minimum upper-bound  $\mathbf{a}^\top \Sigma_M(u) \mathbf{a}$ . The truncation ensures that the pilot estimate can never produce a looser directional bound than the isotropic proxy.

Let  $T_\ell^{\text{pil}}$  be the round at which the pilot stage ends and set

$$\mathcal{H}_\ell^{\text{pil}} := \mathcal{F}_{T_\ell^{\text{pil}}}.$$

Conditional on this history, let  $\mathbf{n} = (n_u : u \in \mathcal{U}_\ell)$  be a proposed vector of fresh estimation-round counts. Define

$$\begin{aligned} R_{\ell,C}^{\text{path}}(\mathbf{n}) &:= \left[ \frac{2\zeta_\ell}{m} \sum_{u \in \mathcal{U}_\ell} \frac{\bar{s}_{\ell,u}^2(\mathbf{a}_{\ell,C,u})}{n_u} \right]^{1/2} \\ &\quad + \frac{2\sigma\zeta_\ell}{3m} \max_{u: \mathbf{a}_{\ell,C,u} \neq 0} \frac{\|\mathbf{a}_{\ell,C,u}\|_2}{n_u}. \end{aligned} \quad (30)$$

For an edge  $e = (u, v) \in \mathcal{E}_\ell$ , put  $q_e := v - u$  and let  $(q_e, 0) \in \mathbb{R}^{2J}$  denote the vector with  $q_e$  in the ordinary-demand block and zero in the weighted block. Define

$$R_{\ell,e}^S(\mathbf{n}) := \left[ \frac{2\zeta_\ell}{m} \left\{ \frac{\bar{s}_{\ell,u}^2((q_e, 0))}{n_u} + \frac{\bar{s}_{\ell,v}^2((q_e, 0))}{n_v} \right\} \right]^{1/2} + \frac{2\sigma\|q_e\|_2\zeta_\ell}{3m} \max \left\{ \frac{1}{n_u}, \frac{1}{n_v} \right\}. \quad (31)$$

**Lemma 5.3** (Isotropic feasibility). *The equal allocation  $n_u = n_\ell^{\text{iso}}$  satisfies*

$$R_{\ell,C}^{\text{path}}(\mathbf{n}) \leq \gamma_\ell \quad \text{for every } C \in \mathcal{C}_\ell^{\text{act}}.$$

*Proof.* By (29),  $\bar{s}_{\ell,u}^2(\mathbf{a}) \leq \sigma^2\|\mathbf{a}\|_2^2$ . For every canonical path,  $\sum_u \|\mathbf{a}_{\ell,C,u}\|_2^2 \leq 2J$  and  $\max_u \|\mathbf{a}_{\ell,C,u}\|_2 \leq \sqrt{J}$ . The two terms in (27) make the variance and bounded-increment components of  $R_{\ell,C}^{\text{path}}$  no larger than  $\gamma_\ell/2$  each.  $\square$

The continuous allocation is chosen by solving

$$\begin{aligned} & \min_{\mathbf{n}} \quad \sum_{u \in \mathcal{U}_\ell} n_u \\ & \text{subject to} \quad R_{\ell,C}^{\text{path}}(\mathbf{n}) \leq \gamma_\ell, \quad C \in \mathcal{C}_\ell^{\text{act}}, \\ & \quad \quad \quad n_u \geq 1, \quad u \in \mathcal{U}_\ell. \end{aligned} \quad (32)$$

A fixed deterministic tie-breaking rule is used if the minimizer is not unique. If  $\mathbf{n}_\ell^*$  is the selected minimizer, the implemented counts are  $n_{\ell,u} := \lceil n_{\ell,u}^* \rceil$ .

**Proposition 5.1** (Convexity and existence of the allocation). *The program in (32) is feasible and convex and admits a minimizer. Rounding every coordinate of a feasible allocation upward preserves feasibility.*

*Proof.* For nonnegative constants  $c_u$ , the map  $\mathbf{n} \mapsto (\sum_u c_u/n_u)^{1/2}$  is convex on the positive orthant. It is the Euclidean norm of the nonnegative vector with coordinates  $\sqrt{c_u/n_u}$ ; each coordinate is convex, and the Euclidean norm is convex and coordinatewise nondecreasing on the nonnegative orthant. The second term in (30) is a maximum of convex functions proportional to  $1/n_u$ . Hence every constraint is convex.

Feasibility follows from Lemma 5.3. The objective has compact sublevel sets on  $[1, \infty)^{|\mathcal{U}_\ell|}$ , so a minimizer exists. Both radius terms are nonincreasing in each coordinate, and therefore rounding upward preserves feasibility.  $\square$

For one fixed path, the leading variance-optimal allocation has a closed form. This calculation isolates the statistical role of the conditional covariance.

**Lemma 5.4** (Oracle allocation along one path). *Fix an active cell  $C$  and let  $\mathcal{I}_C := \{u : \mathbf{a}_{\ell,C,u}^\top \Sigma_M(u) \mathbf{a}_{\ell,C,u} > 0\}$ . Put*

$$s_u := (\mathbf{a}_{\ell,C,u}^\top \Sigma_M(u) \mathbf{a}_{\ell,C,u})^{1/2}.$$

For a total of  $N$  estimation rounds allocated over  $\mathcal{I}_C$ ,

$$\min_{\substack{n_u > 0 \\ \sum_{u \in \mathcal{I}_C} n_u = N}} \sum_{u \in \mathcal{I}_C} \frac{s_u^2}{n_u} = \frac{1}{N} \left( \sum_{u \in \mathcal{I}_C} s_u \right)^2.$$

The minimum is attained by  $n_u = N s_u / \sum_{v \in \mathcal{I}_C} s_v$ .

*Proof.* Cauchy–Schwarz gives

$$\left( \sum_{u \in \mathcal{I}_C} s_u \right)^2 \leq \left( \sum_{u \in \mathcal{I}_C} \frac{s_u^2}{n_u} \right) \left( \sum_{u \in \mathcal{I}_C} n_u \right).$$

Equality holds precisely when  $n_u$  is proportional to  $s_u$ .  $\square$

Bregman curvature and conditional covariance therefore play distinct roles. Bregman curvature determines which tax nodes are needed and how much deterministic reconstruction error a path incurs. Conditional covariance determines how often the selected nodes should be sampled.

### 5.3 Pathwise reconstruction and edge certificates

After solving the allocation program, collect the allocated fresh estimation rounds. Let

$$\hat{\mu}_\ell(u) := \begin{pmatrix} \hat{G}_\ell(u) \\ \hat{D}_\ell(u) \end{pmatrix}$$

be the resulting joint node mean. For an active cell  $C$ , define the *pathwise welfare estimate*

$$\widehat{W}_\ell(C) := x_C^\top \hat{G}_\ell(x_C) - \sum_{(u,v) \in \pi_\ell(x_C)} \langle \hat{D}_\ell(u), v - u \rangle.$$

Equivalently,

$$\widehat{W}_\ell(C) = \sum_{u \in \mathcal{U}_\ell} \langle \mathbf{a}_{\ell,C,u}, \hat{\mu}_\ell(u) \rangle.$$

The ordinary and weighted means at one node are estimated from the same households and are therefore treated jointly rather than as independent quantities.

**Lemma 5.5** (Bregman reconstruction bias). *For every active cell  $C$ ,*

$$\begin{aligned} 0 &\leq W(x_C) - \mathbb{E}_M \left[ \widehat{W}_\ell(C) \mid \mathcal{H}_\ell^{\text{pil}} \right] \\ &\leq b \sum_{(u,v) \in \pi_\ell(x_C)} B_V(v, u) \leq \frac{b J L_V}{2} \gamma_\ell. \end{aligned} \tag{33}$$

*Proof.* Write  $\pi_\ell(x_C) = (u_0, \dots, u_K)$ , with  $u_0 = 0$  and  $u_K = x_C$ . Equation (7) gives  $P(u_{j+1}) - P(u_j) = -\langle D(u_j), u_{j+1} - u_j \rangle + B_P(u_{j+1}, u_j)$ . Summing along the path and using  $P(0) = 0$  shows that the first difference in (33) equals  $\sum_j B_P(u_{j+1}, u_j)$ . Theorem 3.1 bounds this by  $b \sum_j B_V(u_{j+1}, u_j)$ . Finally, (8) and the bound of  $J/\gamma_\ell$  edges of squared length  $\gamma_\ell^2$  give the last inequality.  $\square$

For an epoch edge, define

$$\widehat{S}_\ell(u, v) := \left\langle \widehat{G}_\ell(u) - \widehat{G}_\ell(v), v - u \right\rangle.$$

**Lemma 5.6** (Pilot-split conditional concentration). *On the simultaneous pilot event, conditional on  $\mathcal{H}_\ell^{\text{pil}}$ , the fresh estimation samples satisfy*

$$\left| \widehat{W}_\ell(C) - \mathbb{E}_M \left[ \widehat{W}_\ell(C) \mid \mathcal{H}_\ell^{\text{pil}} \right] \right| \leq R_{\ell, C}^{\text{path}}(\mathbf{n}_\ell) \quad (34)$$

for every active cell and

$$\left| \widehat{S}_\ell(u, v) - S_V(u, v) \right| \leq R_{\ell, (u, v)}^S(\mathbf{n}_\ell) \quad (35)$$

for every epoch edge, simultaneously with conditional probability at least  $1 - \delta_\mu / (L + 1)$ . Consequently, all pilot, path, and edge bounds hold at every epoch with probability at least  $1 - \delta$ .

*Proof.* Conditional on  $\mathcal{H}_\ell^{\text{pil}}$ , the graph, covariance upper bounds, allocation, and path coefficients are fixed. The fresh estimation batches are independent across graph nodes.

For a fixed cell, the exact conditional variance of the centered path estimate is

$$\frac{1}{m} \sum_{u \in \mathcal{U}_\ell} \frac{\mathbf{a}_{\ell, C, u}^\top \Sigma_M(u) \mathbf{a}_{\ell, C, u}}{n_{\ell, u}}.$$

On the pilot event, this is bounded by the variance component of (30). Each centered household contribution has absolute value at most  $2\sigma \|\mathbf{a}_{\ell, C, u}\|_2 / (mn_{\ell, u})$ . A conditional Bernstein inequality (Boucheron et al., 2013) gives (34) with failure probability at most  $2e^{-\zeta_\ell}$ .

For an edge  $e = (u, v)$ , the error in  $\widehat{S}_\ell(u, v)$  is the sum of one ordinary-demand projection at  $u$  and one at  $v$ . Its conditional variance and maximal increment are bounded by the two components of (31). The same Bernstein inequality (Boucheron et al., 2013) gives (35) with failure probability at most  $2e^{-\zeta_\ell}$ .

There are at most  $K_\ell^{\text{grid}}$  active representatives and at most  $JK_\ell^{\text{grid}}$  epoch edges. The definition of  $\zeta_\ell$  and a conditional union bound make the total estimation failure probability at epoch  $\ell$  no greater than  $\delta_\mu / (L + 1)$ . Summing over epochs and combining with Lemma 5.2 and (26) proves the global statement.  $\square$

Put

$$\overline{S}_\ell(u, v) := \widehat{S}_\ell(u, v) + R_{\ell, (u, v)}^S(\mathbf{n}_\ell).$$

For the unrestricted model class, define the directed-edge certificate

$$\bar{c}_\ell^0(u, v) := \min \left\{ \frac{L_V}{2} \|v - u\|_2^2, [\overline{S}_\ell(u, v)]_+ \right\}.$$

If the model supplies a known envelope  $\overline{\varpi}_V(r) \geq \varpi_V(r)$ , define the refined certificate

$$\bar{c}_\ell^\varpi(u, v) := \min \left\{ \frac{L_V}{2} \|v - u\|_2^2, [\bar{S}_\ell(u, v)]_+, \right. \\ \left. \frac{1}{2} [\bar{S}_\ell(u, v)]_+ + \frac{1}{4} \overline{\varpi}_V(\|v - u\|_2) \|v - u\|_2^2 \right\}.$$

**Proposition 5.2** (Validity of the edge certificates). *On the edge confidence event,*

$$B_V(v, u) \leq \bar{c}_\ell^0(u, v). \quad (36)$$

*If a valid Hessian-modulus envelope is available, then*

$$B_V(v, u) \leq \bar{c}_\ell^\varpi(u, v). \quad (37)$$

*In the quadratic case, the refined certificate converges to  $B_V(v, u)$  as the endpoint sample sizes diverge.*

*Proof.* Smoothness gives  $B_V(v, u) \leq L_V \|v - u\|_2^2/2$ . Nonnegativity of the reverse Bregman divergence gives  $B_V(v, u) \leq S_V(u, v) \leq \bar{S}_\ell(u, v)$ . These prove (36).

Proposition 3.8 gives  $B_V(v, u) \leq S_V(u, v)/2 + \varpi_V(\|v - u\|_2) \|v - u\|_2^2/4$ . Replacing both unknown quantities by valid upper bounds proves (37). If  $V$  is quadratic, then  $\varpi_V \equiv 0$  and  $S_V(u, v) = 2B_V(v, u)$ .  $\square$

The broad-class algorithm uses  $\bar{c}_\ell = \bar{c}_\ell^0$ . For each active cell, define the certified pathwise Bregman allowance

$$\bar{\beta}_\ell(C) := b \sum_{(u,v) \in \pi_\ell(x_C)} \bar{c}_\ell(u, v).$$

On the simultaneous confidence event,

$$\sum_{(u,v) \in \pi_\ell(x_C)} B_P(v, u) \leq \bar{\beta}_\ell(C) \leq \frac{bJL_V}{2} \gamma_\ell. \quad (38)$$

## 5.4 Bregman Contrast Pruning

The algorithm starts with all coarsest cells active. At epoch  $\ell$ , it constructs the common recovery graph, collects covariance pilots, solves (32), collects the fresh estimation rounds, and forms the representative lower bound and cell-wide upper bound

$$L_\ell(C) := \widehat{W}_\ell(C) - R_{\ell,C}^{\text{path}}(\mathbf{n}_\ell), \\ U_\ell(C) := \widehat{W}_\ell(C) + R_{\ell,C}^{\text{path}}(\mathbf{n}_\ell) + \bar{\beta}_\ell(C) + L_W \sqrt{J} \gamma_\ell. \quad (39)$$

The lower bound applies to the tested representative. The upper bound also accounts for Bregman reconstruction bias and welfare variation inside the cell.

Let

$$L_\ell^{\max} := \max_{C \in \mathcal{C}_\ell^{\text{act}}} L_\ell(C).$$

The surviving cells are

$$\mathcal{K}_\ell := \{C \in \mathcal{C}_\ell^{\text{act}} : U_\ell(C) \geq L_\ell^{\max}\}.$$

If  $\ell < L$ , their children form the next active collection,

$$\mathcal{C}_{\ell+1}^{\text{act}} := \bigcup_{C \in \mathcal{K}_\ell} \text{Children}(C).$$

At epoch  $L$ , the algorithm recommends  $x_{\hat{C}}$  for any  $\hat{C} \in \arg \max_{C \in \mathcal{K}_L} L_L(C)$ . The full algorithm pseudocode is presented below.

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**Algorithm 1** Bregman Contrast Pruning

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**Require:** Terminal epoch  $L$ , confidence level  $\delta$ , dyadic partition hierarchy  $\{\mathcal{C}_\ell, \mathcal{Q}_\ell\}_{\ell=0}^L$ , and known model bounds  $b, L_V, L_W$ , and  $\sigma^2$ .

**Ensure:** Recommended taxation  $\hat{X}$ .

- 1: Set  $\delta_\Sigma = \delta_\mu = \delta/2$  and initialize  $\mathcal{C}_0^{\text{act}} \leftarrow \mathcal{C}_0$ .
- 2: **for**  $\ell = 0, \dots, L$  **do**
- 3:     For each active cell  $C$ , take its representative taxation  $x_C \in \mathcal{Q}_\ell$ , connect 0 to  $x_C$  by the monotone lattice path  $\pi_\ell(x_C)$ , and merge the paths into the recovery graph  $\mathcal{G}_\ell = (\mathcal{U}_\ell, \mathcal{E}_\ell)$ .
- 4:     Collect an independent covariance pilot at every  $u \in \mathcal{U}_\ell$ , construct the covariance upper bounds  $\bar{\Sigma}_\ell(u)$ , and solve (32) to determine the node-specific estimation counts  $n_{\ell,u}$ .
- 5:     Collect  $n_{\ell,u}$  fresh estimation waves at every  $u \in \mathcal{U}_\ell$  and estimate ordinary and weighted mean demand,  $\hat{G}_\ell(u)$  and  $\hat{D}_\ell(u)$ .
- 6:     For each active cell  $C$ , combine the estimated tax revenue at  $x_C$ , the weighted demand estimates along  $\pi_\ell(x_C)$ , and the certified edge costs to form the representative lower bound  $L_\ell(C)$  and the whole-cell upper bound  $U_\ell(C)$  in (39).
- 7:     Retain the cells whose upper bounds remain competitive:

$$\mathcal{K}_\ell \leftarrow \left\{ C \in \mathcal{C}_\ell^{\text{act}} : U_\ell(C) \geq \max_{C' \in \mathcal{C}_\ell^{\text{act}}} L_\ell(C') \right\}.$$

- 8:     **if**  $\ell < L$  **then**
- 9:         Refine the surviving cells:

$$\mathcal{C}_{\ell+1}^{\text{act}} \leftarrow \bigcup_{C \in \mathcal{K}_\ell} \text{Children}(C).$$

- 10:     **end if**
- 11: **end for**
- 12: Choose

$$\hat{C} \in \arg \max_{C \in \mathcal{K}_L} L_L(C)$$

and return  $\hat{X} \leftarrow x_{\hat{C}}$ .

---

In the main fixed-lattice implementation, small directional Bregman costs tighten cell bounds and may accelerate pruning. They do not alter the candidate edge lengths. A fully sequential extension can use small certified costs to retain longer edges, but that extension is separate from the main algorithm.

## 5.5 Correctness

For a cell  $C$ , define its welfare gap by

$$\Delta_M(C) := W_M^* - \sup_{x \in C} W_M(x).$$

When the environment is fixed, the subscript  $M$  is suppressed. Set

$$C_{\text{BCP}} := 2 + L_W \sqrt{J} + \frac{bJL_V}{2}.$$

**Theorem 5.1** (Correctness of Bregman Contrast Pruning). *With probability at least  $1 - \delta$ , the following statements hold simultaneously for every epoch  $\ell \leq L$ .*

1. *For every active cell,  $L_\ell(C) \leq W(x_C)$  and  $\sup_{x \in C} W(x) \leq U_\ell(C)$ .*
2. *No cell containing a welfare-maximizing taxation is pruned.*
3. *Every active cell satisfying  $\Delta(C) > 2C_{\text{BCP}}\gamma_\ell$  is pruned.*
4. *The final recommendation satisfies*

$$W^* - W(\hat{X}) \leq C_{\text{BCP}}\gamma_L. \quad (40)$$

*Proof.* Work on the simultaneous pilot, path, and edge confidence event. For an active cell  $C$ , put

$$b_\ell(C) := W(x_C) - \mathbb{E}_M \left[ \widehat{W}_\ell(C) \mid \mathcal{H}_\ell^{\text{pil}} \right].$$

Lemma 5.5 and (38) give  $0 \leq b_\ell(C) \leq \bar{\beta}_\ell(C)$ . The path confidence bound therefore yields

$$L_\ell(C) \leq W(x_C) \leq \widehat{W}_\ell(C) + R_{\ell,C}^{\text{path}}(\mathbf{n}_\ell) + \bar{\beta}_\ell(C).$$

For every  $x \in C$ , Lemma 5.1 gives  $W(x) \leq W(x_C) + L_W \sqrt{J} \gamma_\ell$ . This proves the first claim.

The allocation constraint gives  $R_{\ell,C}^{\text{path}}(\mathbf{n}_\ell) \leq \gamma_\ell$ , while (38) gives  $\bar{\beta}_\ell(C) \leq bJL_V\gamma_\ell/2$ . Using both sides of the path confidence interval,

$$U_\ell(C) \leq \sup_{x \in C} W(x) + C_{\text{BCP}}\gamma_\ell. \quad (41)$$

Let  $C^*$  be an active cell containing a maximizer. Its representative satisfies  $W(x_{C^*}) \geq W^* - L_W \sqrt{J} \gamma_\ell$ . The lower side of the path confidence interval gives

$$L_\ell(C^*) \geq W^* - C_{\text{BCP}}\gamma_\ell. \quad (42)$$

Since  $U_\ell(C^*) \geq W^*$  and every lower bound is at most  $W^*$ , an optimal cell is never pruned. If  $\Delta(C) > 2C_{\text{BCP}}\gamma_\ell$ , then (41) and (42) give  $U_\ell(C) < L_\ell(C^*) \leq L_\ell^{\max}$ , so  $C$  is pruned.

At the terminal epoch, the selected cell  $\hat{C}$  satisfies  $L_L(\hat{C}) \geq L_L(C^*)$ . Since  $W(\hat{X}) \geq L_L(\hat{C})$ , (40) follows.  $\square$

Let

$$L_\varepsilon := \min \{ \ell \geq 0 : C_{\text{BCP}}\gamma_\ell \leq \varepsilon \}.$$

If the algorithm is run through epoch  $L_\varepsilon$ , then

$$\mathbb{P}_M \left\{ W^* - W(\hat{X}) \leq \varepsilon \right\} \geq 1 - \delta.$$

The procedure is therefore  $(\varepsilon, \delta)$ -correct.

## 5.6 Sample complexity and fixed-budget regret

The epoch completion times have the required stopping-time property.

**Lemma 5.7** (Epoch completion times). *Set  $T_0 = 0$ . Suppose  $\mathcal{U}_\ell$  is  $\mathcal{F}_{T_\ell}$ -measurable, and use the fixed measurable tie-breaking rule in the allocation program. Then the epoch-start and pilot-completion times are stopping times.*

*Proof.* If  $T$  is a stopping time and  $N$  is a nonnegative integer-valued  $\mathcal{F}_T$ -measurable random variable, then

$$\{T + N \leq t\} = \bigcup_{s=0}^t (\{T = s\} \cap \{N \leq t - s\}) \in \mathcal{F}_t.$$

Apply this observation first to the pilot duration  $n_\ell^{\text{pil}}|\mathcal{U}_\ell|$  and then to the  $\mathcal{H}_\ell^{\text{pil}}$ -measurable estimation duration  $\sum_u n_{\ell,u}$ , proceeding inductively over epochs.  $\square$

The *fixed-confidence stopping time* is

$$\tau_{\varepsilon,\delta} := T_{L_\varepsilon+1} = \sum_{\ell=0}^{L_\varepsilon} \left[ n_\ell^{\text{pil}}|\mathcal{U}_\ell| + \sum_{u \in \mathcal{U}_\ell} n_{\ell,u} \right].$$

Its expectation is the fixed-confidence round complexity. The expected number of household observations is  $m\mathbb{E}_M[\tau_{\varepsilon,\delta}]$ .

**Proposition 5.3** (Isotropic fallback). *The equal allocation  $n_u = n_\ell^{\text{iso}}$  for every  $u \in \mathcal{U}_\ell$  is feasible for (32). Consequently, after rounding,*

$$\sum_{u \in \mathcal{U}_\ell} n_{\ell,u} \leq 2|\mathcal{U}_\ell|n_\ell^{\text{iso}}. \quad (43)$$

Moreover,

$$n_\ell^{\text{pil}} \leq 2n_\ell^{\text{iso}}.$$

*Proof.* By (29),  $\bar{s}_{\ell,u}^2(\mathbf{a}) \leq \sigma^2 \|\mathbf{a}\|_2^2$ . For every canonical path,

$$\sum_{u \in \mathcal{U}_\ell} \|\mathbf{a}_{\ell,C,u}\|_2^2 \leq \|x_C\|_2^2 + \sum_{(u,v) \in \pi_\ell(x_C)} \|v - u\|_2^2 \leq 2J,$$

and  $\max_u \|\mathbf{a}_{\ell,C,u}\|_2 \leq \sqrt{J}$ . The two terms in (27) therefore make the variance and bounded-increment components of every path radius no greater than  $\gamma_\ell/2$ . The equal allocation is feasible.

The continuous minimizer has objective value at most  $|\mathcal{U}_\ell| n_\ell^{\text{iso}}$ . Rounding adds less than  $|\mathcal{U}_\ell|$ , which is no larger than the same quantity because  $n_\ell^{\text{iso}} \geq 1$ . This proves (43). Finally,  $\lceil (n_\ell^{\text{iso}})^{2/3} \rceil \leq 2n_\ell^{\text{iso}}$  and  $\lceil 2/m \rceil \leq 2n_\ell^{\text{iso}}$ .  $\square$

**Lemma 5.8** (Worst-case round count). *There is a constant  $c_J$ , depending only on  $J$ , such that completion through epoch  $L$  uses at most*

$$c_J \left[ \gamma_L^{-J} + \frac{\sigma^2 J}{m} \gamma_L^{-(J+2)} \left\{ \log \left( \frac{c_J(L+1)}{\delta} \right) + J \log \left( \frac{2}{\gamma_L} \right) \right\} \right]$$

rounds.

*Proof.* At epoch  $\ell$ ,  $|\mathcal{U}_\ell| \leq |\mathcal{Q}_\ell| \leq 2^J \gamma_\ell^{-J}$ . Proposition 5.3 bounds the pilot and estimation cost by  $4|\mathcal{U}_\ell| n_\ell^{\text{iso}}$ . The second term in (27) is bounded by a numerical multiple of the first because  $\sigma \sqrt{J}/\gamma_\ell \geq 1$ . Thus  $n_\ell^{\text{iso}}$  is bounded by a constant multiple of  $1 + \sigma^2 J \zeta_\ell / (m \gamma_\ell^2)$ . The sums in  $\gamma_\ell^{-J}$  and  $\gamma_\ell^{-(J+2)}$  are geometric and are bounded by constant multiples of their final terms. The definition of  $\zeta_\ell$  gives the displayed logarithmic factor.  $\square$

**Corollary 5.1** (Fixed-confidence sample complexity). *There is a constant  $C$ , depending only on the fixed model parameters, such that*

$$\sup_{M \in \mathcal{M}} \mathbb{E}_M[\tau_{\varepsilon,\delta}] \leq C \left[ \varepsilon^{-J} + \frac{\sigma^2 J}{m} \varepsilon^{-(J+2)} \log \left( \frac{C}{\delta \varepsilon^J} \right) \right]. \quad (44)$$

*Proof.* Minimality of  $L_\varepsilon$  and  $\gamma_{\ell-1} = 2\gamma_\ell$  imply that  $\gamma_{L_\varepsilon}$  is bounded above and below by fixed multiples of  $\varepsilon$ . Substitute this relation into Lemma 5.8. The resulting bound is pathwise, so taking expectations and then the supremum over  $M$  preserves it.  $\square$

The first term in (44) is the spatial cost of introducing graph nodes. The second is the repeated-sampling cost. The factor  $m$  appears only in the second term because one round introduces one tax policy but contains  $m$  independent household observations.

Let  $\Psi_L(\delta)$  denote the deterministic upper bound in Lemma 5.8. For a fixed budget  $T$ , run Bregman Contrast Pruning with confidence level  $T^{-2}$  through the largest epoch

$$L_T := \max \{L \geq 0 : \Psi_L(T^{-2}) \leq T\}.$$

Let  $\hat{X}_T^{\text{BCP}}$  be the resulting recommendation and define

$$r_T^{\text{BCP}}(M) := W_M^* - W_M(\hat{X}_T^{\text{BCP}}).$$

**Theorem 5.2** (Worst-case regret of Bregman Contrast Pruning). *There is a constant  $C$ , depending only on the fixed model parameters, such that, for all sufficiently large  $T$ ,*

$$\sup_{M \in \mathcal{M}} \mathbb{E}_M [r_T^{\text{BCP}}(M)] \leq C \left[ T^{-1/J} + \left( \frac{\sigma^2 J \log(2T)}{mT} \right)^{1/(J+2)} \right] + \frac{J(1+b)}{T^2}.$$

*Proof.* Set

$$\varepsilon_T := K \left[ T^{-1/J} + \left( \frac{\sigma^2 J \log(2T)}{mT} \right)^{1/(J+2)} \right],$$

where  $K > 0$  is a sufficiently large constant. The first summand gives  $\varepsilon_T^{-J} \leq K^{-J}T$ . The second gives

$$\frac{\sigma^2 J}{m} \varepsilon_T^{-(J+2)} \log(2T) \leq K^{-(J+2)}T.$$

The remaining logarithmic factor in (44) is  $O(\log T)$  because  $\varepsilon_T^{-J} = O(T)$ . Increasing  $K$  if necessary therefore ensures that accuracy  $\varepsilon_T$  is reached within  $T$  rounds.

On the simultaneous confidence event, Theorem 5.1 gives  $r_T^{\text{BCP}}(M) \leq \varepsilon_T$ . The event fails with probability at most  $T^{-2}$ , and Lemma 2.2 bounds regret by  $J(1+b)$  on that event. Taking expectations proves the result.  $\square$

## 5.7 Minimax lower bound and broad-class optimality

The preceding theorem is an upper bound over  $\mathcal{M}$ , and hence over the exact-choice cube class  $\mathcal{M}_{\text{ex}}$ . The hard environments constructed in Proposition 4.2 belong to this same class. It remains to bound the statistical information available about their localized perturbations.

For a budget  $T$ , let  $\mathfrak{A}_T$  be the collection of all possibly randomized, nonanticipating algorithms using at most  $T$  rounds. For  $M \in \mathcal{M}_{\text{ex}}$  and  $\mathcal{A} \in \mathfrak{A}_T$ , define

$$r_T(M, \mathcal{A}) := W_M^* - W_M(\hat{X}_T^{\mathcal{A}}).$$

The minimax expected simple regret is

$$\mathfrak{R}_T^*(\mathcal{M}_{\text{ex}}) := \inf_{\mathcal{A} \in \mathfrak{A}_T} \sup_{M \in \mathcal{M}_{\text{ex}}} \mathbb{E}_M^{\mathcal{A}} [r_T(M, \mathcal{A})].$$

### 5.7.1 Information distance of the structural alternatives

At taxation  $x$ , baseline demand is  $G_0(x) + \xi$  under either latent component. The density of the observed purchase vector is therefore

$$p_0^x(y) = \chi f_A(y - G_0(x)) + (1 - \chi) f_B(y - G_0(x)).$$

Under alternative  $M_k$ , define the local demand displacement

$$s_k(x) := -\frac{\nabla \phi_{k,h}(x)}{1 - \chi}.$$

Component  $A$  is unchanged, while component  $B$  is translated by  $s_k(x)$ . Hence

$$p_k^x(y) = \chi f_A(y - G_0(x)) + (1 - \chi) f_B(y - G_0(x) - s_k(x)).$$

Outside  $R_k$ , the perturbation and its gradient vanish, so  $p_k^x = p_0^x$ . Inside  $R_k$ ,

$$\|s_k(x)\|_2 \leq \frac{\eta K_1}{1 - \chi} h. \quad (45)$$

**Lemma 5.9** (Local information bound). *There are constants  $c_2, h_0 > 0$  such that, for every  $0 < h \leq h_0$ , every alternative  $M_k$ , and every tax  $x \in [0, 1]^J$ ,*

$$\text{KL}(Q_{M_0,1}^x \parallel Q_{M_k,1}^x) \leq c_2 h^2 \mathbb{1}\{x \in R_k\}. \quad (46)$$

Consequently,

$$\text{KL}(Q_{M_0,m}^x \parallel Q_{M_k,m}^x) \leq m c_2 h^2 \mathbb{1}\{x \in R_k\}.$$

*Proof.* The result is immediate outside  $R_k$ . For  $x \in R_k$ , reduce  $h_0$  so that  $\|s_k(x)\|_\infty \leq \nu/4$ . The support of  $p_0^x - p_k^x$  is contained in the union of  $G_0(x) + \text{supp}(f_B)$  and  $G_0(x) + s_k(x) + \text{supp}(f_B)$ . Since  $\text{supp}(f_B) \subseteq [-\nu/4, \nu/4]^J$ , one has  $y - G_0(x) \in [-\nu/2, \nu/2]^J$  throughout this union. By (21),

$$p_k^x(y) \geq \chi c_A$$

wherever  $p_0^x - p_k^x$  is nonzero.

The chi-square upper bound for relative entropy gives

$$\text{KL}(p_0^x \parallel p_k^x) \leq \frac{(1 - \chi)^2}{\chi c_A} \|f_B - f_B(\cdot - s_k(x))\|_2^2.$$

The fundamental theorem of calculus and Minkowski's inequality give

$$\|f_B - f_B(\cdot - s)\|_2 \leq \|s\|_2 \|\nabla f_B\|_2.$$

Combining this with (45) proves (46). The welfare weight equals the same constant  $\bar{\omega}$  in both environments, so adjoining it to the purchase vector does not change relative entropy. The round-level statement follows from (12).  $\square$

### 5.7.2 Adaptive minimax lower bound

**Theorem 5.3** (Structural minimax lower bound). *There is a constant  $c > 0$ , depending only on the fixed parameters of  $\mathcal{M}_{\text{ex}}$ , such that, for all sufficiently large  $T$ ,*

$$\mathfrak{R}_T^*(\mathcal{M}_{\text{ex}}) \geq c \{T^{-1/J} + (mT)^{-1/(J+2)}\}. \quad (47)$$

*Proof.* Fix an arbitrary algorithm  $\mathcal{A} \in \mathfrak{A}_T$ .

For the spatial lower bound, choose  $h = c_s T^{-1/J}$ , where  $c_s > 0$  is sufficiently small that  $N_h \geq 4(T + 1)$ . Under  $M_0$ , one realization of the experiment visits at most  $T$  of the pairwise disjoint regions, and the final recommendation belongs to at most one additional region. Hence

$$\sum_{k=1}^{N_h} \mathbb{P}_{M_0}^{\mathcal{A}} \left( \{\exists t \leq T : X_t \in R_k\} \cup \{\hat{X}_T^{\mathcal{A}} \in R_k\} \right) \leq T + 1. \quad (48)$$

There is therefore an index  $k$  for which the probability in (48) is at most  $1/4$ .

The observation laws under  $M_0$  and  $M_k$  agree at every taxation outside  $R_k$ . Couple the two experiments using the same algorithmic randomization and the same household observations until the first tax in  $R_k$  is selected. On the event that the algorithm never enters  $R_k$ , the complete histories and final recommendations coincide. Thus, with probability at least  $3/4$  under  $M_k$ , the algorithm recommends outside  $R_k$ . Every such taxation has welfare at most  $w_0$ , whereas the optimal welfare under  $M_k$  is at least  $w_0 + c_1 h$ . Consequently,

$$\sup_{M \in \mathcal{M}_{\text{ex}}} \mathbb{E}_M^{\mathcal{A}} [r_T(M, \mathcal{A})] \geq c T^{-1/J}.$$

For the statistical lower bound, choose  $h = c_i (mT)^{-1/(J+2)}$ . Define

$$N_k(T) := \sum_{t=1}^T \mathbb{1}\{X_t \in R_k\}, \quad q_k := \mathbb{P}_{M_0}^{\mathcal{A}} \{\hat{X}_T^{\mathcal{A}} \in R_k\}.$$

Since the regions are pairwise disjoint,

$$\sum_{k=1}^{N_h} \left\{ \frac{\mathbb{E}_{M_0}^{\mathcal{A}} [N_k(T)]}{T} + q_k \right\} \leq 2.$$

Hence some  $k$  satisfies

$$\mathbb{E}_{M_0}^{\mathcal{A}} [N_k(T)] \leq \frac{2T}{N_h}, \quad q_k \leq \frac{2}{N_h}.$$

Let  $\mathbb{P}_M^{\mathcal{A}, T}$  denote the law of the complete experimental history through round  $T$ . The adaptive chain rule, predictability of the selected taxes, and Lemma 5.9 give

$$\begin{aligned} \text{KL} \left( \mathbb{P}_{M_0}^{\mathcal{A}, T} \parallel \mathbb{P}_{M_k}^{\mathcal{A}, T} \right) &= \mathbb{E}_{M_0}^{\mathcal{A}} \left[ \sum_{t=1}^T \text{KL} (Q_{M_0, m}^{X_t} \parallel Q_{M_k, m}^{X_t}) \right] \\ &\leq m c_2 h^2 \mathbb{E}_{M_0}^{\mathcal{A}} [N_k(T)] \\ &\leq \frac{2c_2}{c_0} m T h^{J+2}. \end{aligned}$$

Choose  $c_i$  sufficiently small that the final expression is at most  $1/8$ . Pinsker's inequality (Tsybakov, 2009) then gives total variation at most  $1/4$ . For all sufficiently large  $T$ , one also has  $2/N_h \leq 1/4$ . Therefore,

$$\mathbb{P}_{M_k}^{\mathcal{A}} \{\hat{X}_T^{\mathcal{A}} \in R_k\} \leq q_k + \left\| \mathbb{P}_{M_0}^{\mathcal{A}, T} - \mathbb{P}_{M_k}^{\mathcal{A}, T} \right\|_{\text{TV}} \leq \frac{1}{2}.$$

With probability at least  $1/2$ , the recommendation lies outside  $R_k$  and incurs regret at least  $c_1 h$ . Thus

$$\sup_{M \in \mathcal{M}_{\text{ex}}} \mathbb{E}_M^{\mathcal{A}} [r_T(M, \mathcal{A})] \geq c(mT)^{-1/(J+2)}.$$

Both lower bounds hold for every algorithm. Taking the infimum over  $\mathcal{A}$  and using  $\max\{r, s\} \geq (r + s)/2$  proves (47).  $\square$

### 5.7.3 Unified minimax rate

**Theorem 5.4** (Structural minimax regret). *There are constants  $c, C > 0$ , depending only on the fixed parameters of  $\mathcal{M}_{\text{ex}}$ , such that, for all sufficiently large  $T$ ,*

$$\begin{aligned} c \{T^{-1/J} + (mT)^{-1/(J+2)}\} &\leq \mathfrak{R}_T^*(\mathcal{M}_{\text{ex}}) \\ &\leq \sup_{M \in \mathcal{M}_{\text{ex}}} \mathbb{E}_M [r_T^{\text{BCP}}(M)] \\ &\leq C \left[ T^{-1/J} + \left( \frac{\sigma^2 J \log(2T)}{mT} \right)^{1/(J+2)} \right] + \frac{J(1+b)}{T^2}. \end{aligned}$$

For fixed  $J$ , fixed  $m$ , and fixed model constants,

$$\mathfrak{R}_T^*(\mathcal{M}_{\text{ex}}) = \tilde{\Theta}(T^{-1/(J+2)}).$$

*Proof.* The lower bound is Theorem 5.3. The upper bound follows from Theorem 5.2 because  $\mathcal{M}_{\text{ex}} \subseteq \mathcal{M}$ .  $\square$

The theorem separates a spatial search cost from a statistical information cost. The term  $T^{-1/J}$  reflects the fact that one round can test only one tax vector. The term  $(mT)^{-1/(J+2)}$  combines the  $h^{-J}$  possible locations of the welfare improvement with the  $mh^2$  information obtained from one round inside its region.

If an algorithm is  $\varepsilon_T$ -optimal with probability at least  $1 - \delta_T$ , then Lemma 2.2 gives

$$\mathbb{E}_M[r_T] \leq \varepsilon_T + J(1+b)\delta_T.$$

In a finite tax set with a fixed positive welfare gap, exact identification can have exponentially small error probability. On the continuous domain, the approximation term remains polynomial because the location of the optimum must be resolved increasingly finely.

**Comparison with uniform sampling.** A uniform procedure may use the same dyadic lattice and pathwise estimator but decline to prune. At target accuracy  $\varepsilon$ , it samples all  $O(\varepsilon^{-J})$  lattice nodes and repeats each node sufficiently often to control statistical error. Its fixed-confidence round complexity satisfies

$$T_{\text{unif}}(\varepsilon, \delta) = O \left[ \varepsilon^{-J} + \frac{\sigma^2 J}{m} \varepsilon^{-(J+2)} \log \left( \frac{C}{\delta \varepsilon^J} \right) \right].$$

Theorem 5.4 shows that its broad-class polynomial exponent is unavoidable. The advantage of pruning must therefore arise through favorable instance-dependent structure rather than a better unrestricted minimax exponent.

## 5.8 Localized recovery complexity

The graph produced by the algorithm is random. A deterministic, model-dependent envelope gives a cleaner description of favorable instances. For  $M \in \mathcal{M}$ , define

$$\mathcal{N}_\ell(M) := \left\{ C \in \mathcal{C}_\ell : W_M^* - \sup_{x \in C} W_M(x) \leq 2C_{\text{BCP}}\gamma_\ell \right\}.$$

Set  $\bar{\mathcal{C}}_0(M) := \mathcal{C}_0$ , and for  $\ell \geq 1$  define

$$\bar{\mathcal{C}}_\ell(M) := \bigcup_{C \in \mathcal{N}_{\ell-1}(M)} \text{Children}(C), \quad \bar{\mathcal{U}}_\ell(M) := \bigcup_{C \in \bar{\mathcal{C}}_\ell(M)} \text{nodes}(\pi_\ell(x_C)).$$

**Lemma 5.10** (Localization of the active graph). *On the correctness event,*

$$\mathcal{C}_\ell^{\text{act}} \subseteq \bar{\mathcal{C}}_\ell(M), \quad \mathcal{U}_\ell \subseteq \bar{\mathcal{U}}_\ell(M)$$

for every  $\ell \geq 1$ .

*Proof.* Every active cell at epoch  $\ell$  has a parent that survived epoch  $\ell - 1$ . Theorem 5.1 eliminates every parent whose welfare gap exceeds  $2C_{\text{BCP}}\gamma_{\ell-1}$ . The first inclusion follows, and the graph inclusion follows by taking unions of the corresponding paths.  $\square$

Define the *localized recovery count* by

$$\bar{N}_M(\varepsilon) := \sum_{\ell=0}^{L_\varepsilon} |\bar{\mathcal{U}}_\ell(M)|.$$

The environment has *localized recovery exponent*  $d_M > 0$  if there is  $C_M < \infty$  such that

$$\bar{N}_M(\varepsilon) \leq C_M \varepsilon^{-d_M}$$

for every sufficiently small  $\varepsilon$ . This exponent depends on the environment together with the chosen hierarchy and path architecture; it is not an intrinsic dimension of welfare alone.

**Theorem 5.5** (Localized simple regret). *Suppose  $M$  has localized recovery exponent  $d_M$ . Then*

$$\begin{aligned} \mathbb{E}_M [r_T^{\text{BCP}}(M)] &\leq C \left[ \left( \frac{C_M}{T} \right)^{1/d_M} \right. \\ &\quad \left. + \left( \frac{C_M \sigma^2 J \log(CT)}{mT} \right)^{1/(d_M+2)} \right] + \frac{J(1+b)}{T^2}. \end{aligned} \quad (49)$$

*Proof.* At target accuracy  $\varepsilon$ , on the correctness event, Lemma 5.10 bounds the total number of node occurrences by  $C_M \varepsilon^{-d_M}$ . Applying the isotropic fallback at those nodes gives, up to fixed constants and logarithmic factors, a round count of

$$C_M \varepsilon^{-d_M} + \frac{C_M \sigma^2 J}{m} \varepsilon^{-(d_M+2)} \log \left( \frac{C}{\delta \varepsilon} \right). \quad (50)$$

Choose  $\delta = T^{-2}$  and choose  $\varepsilon$  to be a sufficiently large constant multiple of the two terms displayed in (49). Both terms in (50) are then at most fixed fractions of  $T$ . The failure event is controlled by Lemma 2.2.  $\square$

The notation  $\tilde{O}$  suppresses logarithmic factors in the experimental budget. For fixed  $m$  and model constants, Theorem 5.5 gives  $\tilde{O}(T^{-1/(d_M+2)})$ .

## 5.9 Quadratic welfare growth

For a finite representative set  $\mathcal{R}_\ell \subseteq \mathcal{Q}_\ell$ , define its coordinate widths by

$$w_{\ell,j} := \max_{x \in \mathcal{R}_\ell} x_j - \min_{x \in \mathcal{R}_\ell} x_j.$$

**Lemma 5.11** (Shared-path node count). *The union of canonical paths to  $\mathcal{R}_\ell$  contains at most*

$$(1 + \gamma_\ell^{-1}) \sum_{j=1}^J \prod_{k=1}^{j-1} \left(1 + \frac{w_{\ell,k}}{\gamma_\ell}\right)$$

*nodes, where the product over the empty set equals one.*

*Proof.* The segment of a canonical path on which coordinate  $j$  changes is determined by the first  $j - 1$  coordinates of its endpoint. There are at most  $\prod_{k < j} (1 + w_{\ell,k}/\gamma_\ell)$  distinct such prefixes, and each prefix uses at most  $1 + \gamma_\ell^{-1}$  nodes while coordinate  $j$  changes. Summing over  $j$  proves the result; repeated nodes are only overcounted.  $\square$

Suppose welfare has a unique maximizer and satisfies the *quadratic-growth condition*

$$W^* - W(x) \geq \frac{\mu}{2} \|x - x^*\|_2^2, \quad x \in \mathcal{X}, \quad (51)$$

for some  $\mu > 0$ .

**Theorem 5.6** (Improved BCP rate under quadratic welfare growth). *Under (51), there is  $C_\mu < \infty$  such that*

$$|\overline{\mathcal{U}}_\ell(M)| \leq C_\mu \gamma_\ell^{-(J+1)/2}.$$

*Consequently,  $d_M \leq (J + 1)/2$  and*

$$\begin{aligned} \mathbb{E}_M [r_T^{\text{BCP}}(M)] &\leq C'_\mu \left[ T^{-2/(J+1)} \right. \\ &\quad \left. + \left( \frac{\sigma^2 J \log T}{mT} \right)^{2/(J+5)} \right] + \frac{J(1+b)}{T^2}. \end{aligned}$$

*Proof.* Take a cell in  $\overline{\mathcal{C}}_\ell(M)$ . Its parent  $C^-$  satisfies  $W^* - \sup_{x \in C^-} W(x) \leq 2C_{\text{BCP}}\gamma_{\ell-1}$ . Let  $y \in C^-$  attain the supremum. Quadratic growth gives

$$\|y - x^*\|_2 \leq \left( \frac{4C_{\text{BCP}}\gamma_{\ell-1}}{\mu} \right)^{1/2}.$$

Every child representative is at most  $\sqrt{J} \gamma_{\ell-1}$  from  $y$ . Thus all representatives in  $\bar{\mathcal{C}}_\ell(M)$  lie in a ball of radius  $C\sqrt{\gamma_\ell}$  around  $x^*$ , and each coordinate width is at most  $C\sqrt{\gamma_\ell}$ .

Lemma 5.11 then gives

$$|\bar{\mathcal{U}}_\ell(M)| \leq C\gamma_\ell^{-1} \sum_{j=1}^J \gamma_\ell^{-(j-1)/2} \leq C_\mu \gamma_\ell^{-(J+1)/2}.$$

The rate follows from Theorem 5.5 with  $d_M = (J+1)/2$ .  $\square$

The exponent  $(J+1)/2$  reflects both localization of the active cells and the overhead of connecting them to the reference taxation through canonical coordinate paths. It is therefore a property of the environment together with the chosen recovery architecture.

**Corollary 5.2** (Homogeneous quadratic specialization). *Suppose  $V$  is quadratic,  $H \succeq \kappa I_J$  for some  $\kappa > 0$ , and  $\omega(\theta) = \bar{\omega} < 2$  almost surely. Then (51) holds with  $\mu = (2 - \bar{\omega})\kappa$ , and Theorem 5.6 applies.*

*Proof.* Proposition 3.3 gives  $\nabla^2 W = -(2 - \bar{\omega})H \preceq -(2 - \bar{\omega})\kappa I_J$ . Thus  $W$  is  $(2 - \bar{\omega})\kappa$ -strongly concave. Strong concavity and the first-order condition at the constrained maximizer imply (51).  $\square$

The rank-factorization theorem in Section 3.1 suggests a further implementation in which nonlinear search is conducted on the range of the substitution matrix and the null-space component is treated as affine. A formal rank-adapted regret theorem would additionally require a tax domain compatible with those coordinates and either known or consistently estimated subspaces.

## 6 Discussion

I formulate a sequential experiment for learning an approximately welfare-maximizing vector of commodity taxes under households with quasilinear utilities. The policymaker observes ordinary and weighted household demand rather than social welfare itself. Bregman Contrast Pruning uses the conservative structure of demand to construct welfare bounds along directed paths, eliminate tax regions that cannot contain an optimum, and refine the surviving regions. The main practical implication from BCP is that learning an approximately optimal taxation does not require precise information about every possible tax. Sampling can instead be directed toward the policy comparisons that could induce distinctly different recommended taxations.

Over the exact-choice model class including the hard family, the minimax expected simple regret is  $T^{-1/(J+2)}$ . For fixed cohort size and commodity dimension, this regret decay has the same broad-class polynomial exponent as uniform sampling. The result does not imply that pruning is redundant. It shows that the unrestricted class contains environments in which almost every tax region remains statistically plausible until fine spatial resolution is reached. Instead, the advantage of BCP is instance-dependent adaptation. Bregman geometry determines which taxation nodes are needed and how much deterministic extrapolation error is incurred along their paths. Conditional covariance determines how many repeated samples should be allocated to those nodes. Under quadratic welfare growth, the active region contracts around the optimum and the canonical recovery graph has exponent at most  $(J + 1)/2$ , producing expected regret of order  $T^{-2/(J+5)}$  up to logarithmic factors. Rank factorization (Theorem 3.2) suggests that the regret bound could be improved under economic conditions that imply a rank-deficient demand field. However, a formal analysis of such economic conditions is left open.

Bregman Contrast Pruning is decision focused rather than field-estimation focused. Once a region is certified to be suboptimal, additional precision within that region cannot improve the final tax recommendation. This provides a plausible advantage over procedures requiring uniform estimation of the full welfare field, although a formal separation remains open.

Several assumptions are economically substantive. Quasilinearity excludes income effects and assumes that producers pass the full burden of the taxation to consumers. Additionally, the repeated cross-sections design of BCP disallows dynamic household responses such as anticipation of future taxation. Extending the method to income effects, equilibrium price responses, spillovers, or uncertain welfare weights may require a different economic setting.

The minimax lower bound imposes some additional restrictions. It uses continuous, divisible purchase quantities (e.x. a household can buy 20.3 ounces of peanut butter) and continuous preferences (households weakly prefer 20.4 ounces of peanut butter to 20.3 ounces). The purpose of these specifications is to permit the construction of purchase distributions that overlap substantially even though their welfare-maximizing tax regions differ. The resulting lower bound does not immediately apply to a setting in which every commodity purchase is binary or indivisible. Establishing a comparable lower bound for discrete-choice demand would require a new construction of the household model class.

Taken together, the results support a design principle for policy experimentation towards an optimal taxation. When social welfare is not directly observed, experiments should be organized around unresolved policy comparisons. Practically, adopting this principle means concentrating samples on tax regions that could still improve welfare and allocating repeated observations to the directions whose implied social welfare remain statistically uncertain according to the observation noise and which have high extrapolation error according to the demand curvature. Bregman Contrast Pruning incorporates this principle by using the geometry of the demand setting and the covariance of household responses to direct data collection toward the final policy decision.

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